

IR-9 Coordination Compounds

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IR-9.1 INTRODUCTION

IR-9.1.1 **General**

This Chapter presents the definitions and rules necessary for formulating and naming coordination compounds. Key terms such as coordination entity, coordination polyhedron, coordination number, chelation and bridging ligands are first defined and the role of additive nomenclature explained (see also Chapter IR-7).

These definitions are then used to develop rules for writing the names and formulae of coordination compounds. The rules allow the composition of coordination compounds to be described in a way that is as unambiguous as possible. The names and formulae provide information about the nature of the central atom, the ligands that are attached to it, and the overall charge on the structure.

Stereochemical descriptors are then introduced as a means of identifying or distinguishing between the diastereoisomeric or enantiomeric structures that may exist for a compound of any particular composition.

The description of the configuration of a coordination compound requires first that the coordination geometry be specified using a polyhedral symbol (Section IR-9.3.2.1). Once this is done the relative positions of the ligands around the coordination polyhedron are specified using the configuration index (Section IR-9.3.3). The configuration index is a sequence of ligand priority numbers produced by following rules specific to each coordination geometry. If required, the chirality of a coordination compound can be described, again using ligand priority numbers (Section IR-9.3.4). The ligand priority numbers used in these descriptions are based on the chemical composition of the ligands. A detailed description of the rules by which they are obtained is provided in Section P-91 of Ref. 1, but an outline is given in Section IR-9.3.5.

IR-9.1.2 **Definitions**IR-9.1.2.1 *Background*

The development of coordination theory and the identification of a class of compounds called coordination compounds began with the historically significant concepts of primary and secondary valence.

Primary valencies were obvious from the stoichiometries of simple compounds such as NiCl_2 , $\text{Fe}_2(\text{SO}_4)_3$ and PtCl_2 . However, new materials were frequently observed when other, independently stable substances, *e.g.* H_2O , NH_3 or KCl , were added to these simple compounds giving, for example, $\text{NiCl}_2 \cdot 4\text{H}_2\text{O}$, $\text{Co}_2(\text{SO}_4)_3 \cdot 12\text{NH}_3$ or $\text{PtCl}_2 \cdot 2\text{KCl}$. Such species were called complex compounds, in recognition of the stoichiometric complications they represented, and were considered characteristic of certain metallic elements. The number of species considered to be added to the simple compounds gave rise to the concept of secondary valence.

Recognition of the relationships between these complex compounds led to the formulation of coordination theory and the naming of coordination compounds using additive nomenclature. Each coordination compound either is, or contains, a coordination entity (or complex) that consists of a central atom to which other groups are bound.

While these concepts have usually been applied to metal compounds, a wide range of other species can be considered to consist of a central atom or central atoms to which a number of other groups are bound. The application of additive nomenclature to such species is briefly described and exemplified in Chapter IR-7, and abundantly exemplified for inorganic acids in Chapter IR-8.

IR-9.1.2.2 *Coordination compounds and the coordination entity*

A coordination compound is any compound that contains a coordination entity. A coordination entity is an ion or neutral molecule that is composed of a central atom, usually that of a metal, to which is attached a surrounding array of other atoms or groups of atoms, each of which is called a ligand. Classically, a ligand was said to satisfy either a secondary or a primary valence of the central atom and the sum of these valencies (often equal to the number of ligands) was called the *coordination number* (see Section IR-9.1.2.6). In formulae, the coordination entity is enclosed in square brackets whether it is charged or uncharged (see Section IR-9.2.3.2).

Examples:

1. $[\text{Co}(\text{NH}_3)_6]^{3+}$
2. $[\text{PtCl}_4]^{2-}$
3. $[\text{Fe}_3(\text{CO})_{12}]$

IR-9.1.2.3 *Central atom*

The central atom is the atom in a coordination entity which binds other atoms or groups of atoms (ligands) to itself, thereby occupying a central position in the coordination entity. The central atoms in $[\text{NiCl}_2(\text{H}_2\text{O})_4]$, $[\text{Co}(\text{NH}_3)_6]^{3+}$ and $[\text{PtCl}_4]^{2-}$ are nickel, cobalt and platinum, respectively. In general, a name for a (complicated) coordination entity will be more easily produced if more central atoms are chosen (see Section IR-9.2.5) and the connectivity of the structure is indicated using the kappa convention (see Section IR-9.2.4.2).

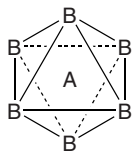
IR-9.1.2.4 *Ligands*

The ligands are the atoms or groups of atoms bound to the central atom. The root of the word is often converted into other forms, such as to ligate, meaning to coordinate as a ligand, and the derived participles, ligating and ligated. The terms ‘ligating atom’ and ‘donor atom’ are used interchangeably.

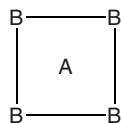
IR-9.1.2.5 *Coordination polyhedron*

It is standard practice to regard the ligand atoms directly attached to the central atom as defining a coordination polyhedron (or polygon) about the central atom. Thus $[\text{Co}(\text{NH}_3)_6]^{3+}$ is an octahedral ion and $[\text{PtCl}_4]^{2-}$ is a square planar ion. In such cases, the coordination number will be equal to the number of vertices in the coordination polyhedron. This may not hold true in cases where one or more ligands coordinate to the central atom through two or more contiguous atoms. It may hold if the contiguous atoms are treated as a single ligand occupying one vertex of the coordination polyhedron.

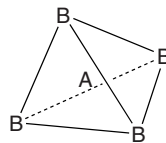
Examples:



1. octahedral
coordination
polyhedron



2. square planar
coordination
polygon



3. tetrahedral
coordination
polyhedron

IR-9.1.2.6 Coordination number

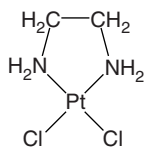
For coordination compounds, the coordination number equals the number of σ -bonds between ligands and the central atom. Note that where both σ - and π -bonding occurs between the ligating atom and the central atom, *e.g.* with ligands such as CN^- , CO , N_2 and PMe_3 , the π -bonds are not considered in determining the coordination number.

IR-9.1.2.7 Chelation

Chelation involves coordination of more than one non-contiguous σ -electron pair donor atom from a given ligand to the same central atom. The number of such ligating atoms in a single chelating ligand is indicated by the adjectives bidentate², tridentate, tetradentate, pentadentate, *etc.* (see Table IV* for a list of multiplicative prefixes). The number of donor atoms from a given ligand attached to the same central atom is called the denticity.

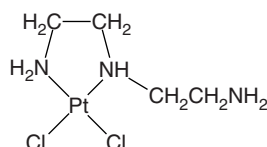
Examples:

1.



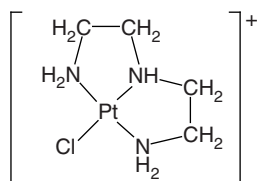
bidentate chelation

2.



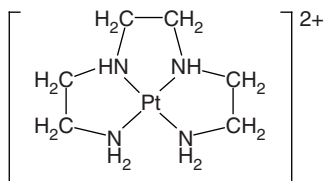
bidentate chelation

3.



tridentate chelation

4.



tetradentate chelation

The cyclic structures formed when more than one donor atom from the same ligand is bound to the central atom are called chelate rings, and the process of coordination of these donor atoms is called chelation.

* Tables numbered with a Roman numeral are collected together at the end of this book.

If a potentially bidentate ligand, such as ethane-1,2-diamine, coordinates to two metal ions, it does *not* chelate but coordinates in a monodentate fashion to each metal ion, forming a connecting link or bridge.

Example:



Alkenes, arenes and other unsaturated molecules attach to central atoms, using some or all of their multiply bonded atoms, to give organometallic complexes. While there are many similarities between the nomenclature of coordination and organometallic compounds, the latter differ from the former in clearly definable ways. Organometallic complexes are therefore treated separately in Chapter IR-10.

IR-9.1.2.8

Oxidation state

The oxidation state of a central atom in a coordination entity is defined as the charge it would bear if all the ligands were removed along with the electron pairs that were shared with the central atom. It is represented by a Roman numeral. It must be emphasized that oxidation state is an index derived from a simple and formal set of rules (see also Sections IR-4.6.1 and IR-5.4.2.2) and that it is not a direct indicator of electron distribution. In certain cases, the formalism does not give acceptable central atom oxidation states. Because of such ambiguous cases, the net charge on the coordination entity is preferred in most nomenclature practices. The following examples illustrate the relationship between the overall charge on a coordination entity, the number and charges of ligands, and the derived central atom oxidation state.

	<i>Formula</i>	<i>Ligands</i>	<i>Central atom oxidation state</i>
1.	$[\text{Co}(\text{NH}_3)_6]^{3+}$	6 NH_3	III
2.	$[\text{CoCl}_4]^{2-}$	4 Cl^-	II
3.	$[\text{MnO}_4]^-$	4 O^{2-}	VII
4.	$[\text{MnFO}_3]$	3 O^{2-} + 1 F^-	VII
5.	$[\text{Co}(\text{CN})_5\text{H}]^{3-}$	5 CN^- + 1 H^-	III
6.	$[\text{Fe}(\text{CO})_4]^{2-}$	4 CO	-II

IR-9.1.2.9

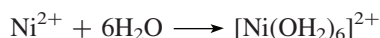
Coordination nomenclature: an additive nomenclature

When coordination theory was first developed, coordination compounds were considered to be formed by addition of independently stable compounds to a simple central compound. They were therefore named on the basis of an additive principle, where the names of the added compounds and the central simple compound were combined. This principle remains the basis for naming coordination compounds.

The name is built up around the central atom name, just as the coordination entity is built up around the central atom.

Example:

1. Addition of ligands to a central atom:



Addition of ligand names to a central atom name:

hexaaquanickel(II)

This nomenclature then extends to more complicated structures where central atoms (and their ligands) are added together to form polynuclear species from mononuclear building blocks. Complicated structures are usually more easily named by treating them as polynuclear species (see Section IR-9.2.5).

IR-9.1.2.10 *Bridging ligands*

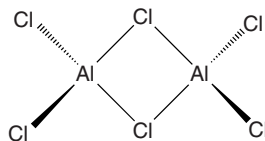
In polynuclear species a ligand can also act as a bridging group, by forming bonds to two or more central atoms simultaneously. Bridging is indicated in names and formulae by adding the symbol μ as a prefix to the ligand formula or name (see Section IR-9.2.5.2).

Bridging ligands link central atoms together to produce coordination entities having more than one central atom. The number of central atoms joined into a single coordination entity by bridging ligands or direct bonds between central atoms is indicated by using the terms dinuclear, trinuclear, tetranuclear, *etc.*

The bridging index is the number of central atoms linked by a particular bridging ligand (see Section IR-9.2.5.2). Bridging can be through one atom or through a longer array of atoms.

Example:

- 1.



di- μ -chlorido-tetrachlorido-1 κ^2 Cl,2 κ^2 Cl-dialuminium

IR-9.1.2.11 *Metal-metal bonds*

Simple structures that contain a metal-metal bond are readily described using additive nomenclature (see Section IR-9.2.5.3), but complications arise for structures that involve three or more central atoms. Species that contain such clusters of central atoms are treated in Sections IR-9.2.5.6 and IR-9.2.5.7.

Examples:

1. $[\text{Br}_4\text{ReReBr}_4]^{2+}$
bis(tetrabromidorhenium)(Re—Re)(2+)
2. $[(\text{OC})_5^1\text{Re}^2\text{Co}(\text{CO})_4]$
nonacarbonyl-1 κ^5 C,2 κ^4 C-rheniumcobalt(Re—Co)

IR-9.2 DESCRIBING THE CONSTITUTION OF COORDINATION COMPOUNDS

IR-9.2.1 General

Three main methods are available for describing the constitution of compounds: one can draw structures, write names or write formulae. A drawn structure contains information about the structural components of the molecule as well as their stereochemical relationships. Unfortunately, such structures are not usually suitable for inclusion in text. Names and formulae are therefore used to describe the constitution of a compound.

The name of a coordination compound provides detailed information about the structural components present. However, it is important that the name can be easily interpreted unambiguously. For that reason, there should be rules that define how the name is constructed. The following sections detail these rules and provide examples of their use.

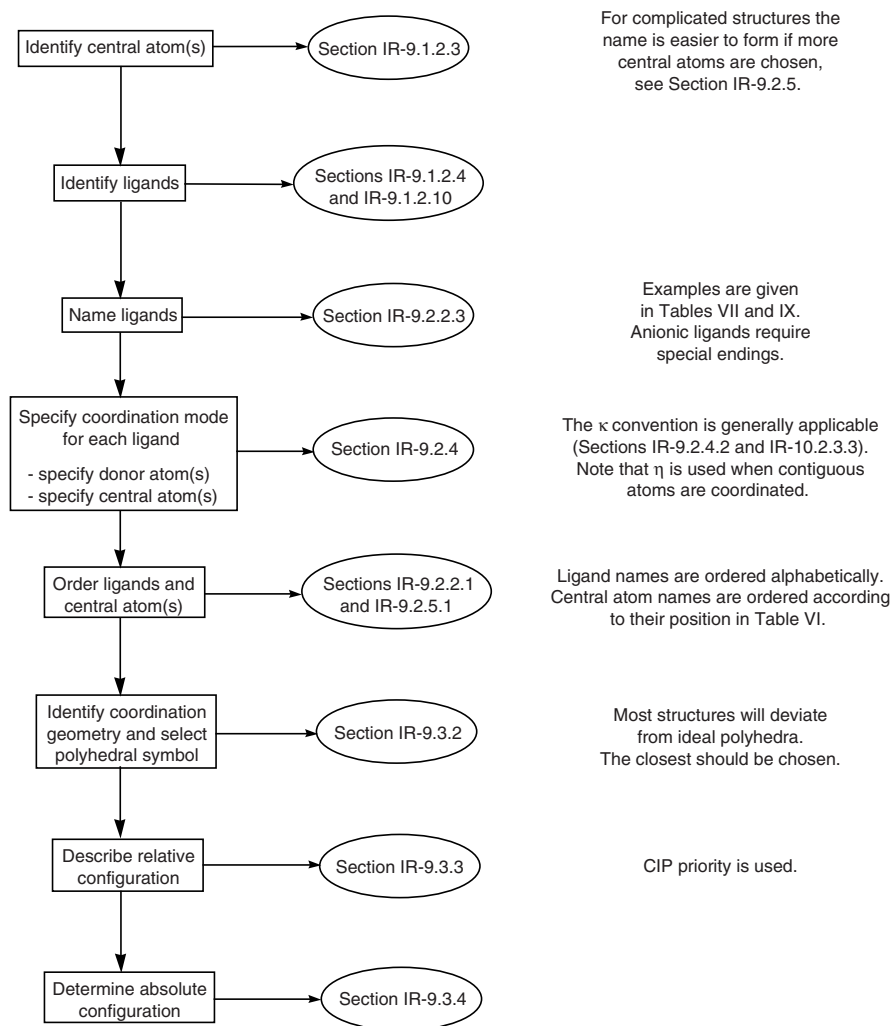


Figure IR-9.1 Stepwise procedure for naming coordination compounds.

The flowchart shown in Figure IR-9.1 illustrates a general procedure for producing a name for a coordination compound. Sections containing the detailed rules, guidelines and examples relevant to each stage of the procedure are indicated.

The name of a compound can, however, be rather long and its use may be inconvenient. In such circumstances a formula provides a shorthand method of representing the compound. Rules are provided in order to make the use of formulae more straightforward. It should be noted that, because of their abbreviated form, it is often not possible to provide as much information about the structure of a compound in its formula as can be provided by its name.

IR-9.2.2 Names of coordination compounds

The systematic names of coordination entities are derived by following the principles of additive nomenclature, as outlined in Chapter IR-7. Thus, the groups that surround the central atom or structure must be identified in the name. They are listed as prefixes to the name of the central atom (see Section IR-9.2.2.1) along with any appropriate multipliers (see Section IR-9.2.2.2). These prefixes are usually derived in a simple way from the ligand names (see Section IR-9.2.2.3). Names of anionic coordination entities are furthermore given the ending 'ate'.

IR-9.2.2.1 Sequences of ligands and central atoms within names

The following general rules are used when naming coordination compounds:

- (i) ligand names are listed before the name(s) of the central atom(s),
- (ii) no spaces are left between parts of the name that refer to the same coordination entity,
- (iii) ligand names are listed in alphabetical order (multiplicative prefixes indicating the number of ligands are not considered in determining that order),
- (iv) the use of abbreviations in *names* is discouraged.

Examples:

1. $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
pentaamminechloridocobalt(2+) chloride
2. $[\text{AuXe}_4]^{2+}$
tetraxenonidogold(2+)

Additional rules which apply to polynuclear compounds are dealt with in Section IR-9.2.5.

IR-9.2.2.2 Number of ligands in a coordination entity

Two kinds of multiplicative prefix are available for indicating the number of each type of ligand within the name of the coordination entity (see Table IV).

- (i) Prefixes di, tri, *etc.* are generally used with the names of simple ligands. Enclosing marks are not required.
- (ii) Prefixes bis, tris, tetrakis, *etc.* are used with complex ligand names and in order to avoid ambiguity. Enclosing marks (the nesting order of which is described in Section IR-2.2) must be placed around the multiplicand.

For example, one would use diammine for $(\text{NH}_3)_2$, but bis(methylamine) for $(\text{NH}_2\text{Me})_2$, to make a distinction from dimethylamine. There is no elision of vowels or use of a hyphen, *e.g.* in tetraammine and similar names.

IR-9.2.2.3 *Representing ligands in names*

Systematic and alternative names for some common ligands are given in Tables VII and IX. Table VII contains the names of common organic ligands whereas Table IX contains the names of other simple molecules and ions that may act as ligands. The general features are as follows:

- (i) Names of anionic ligands, whether inorganic or organic, are modified to end in 'o'. In general, if the anion name ends in 'ide', 'ite' or 'ate', the final 'e' is replaced by 'o', giving 'ido', 'ito' and 'ato', respectively. In particular, alcoholates, thiolates, phenolates, carboxylates, partially dehydrated amines, phosphanes, *etc.* are in this category. Also, it follows that halide ligands are named fluorido, chlorido, bromido and iodido, and coordinated cyanide is named cyanido.

In its complexes, except for those of molecular hydrogen, hydrogen is always treated as anionic. 'Hydrido' is used for hydrogen coordinating to all elements including boron.³

- (ii) Names of neutral and cationic ligands, including organic ligands,⁴ are used without modification (even if they carry the endings 'ide', 'ite' or 'ate'; see Examples 8 and 14 below).
- (iii) Enclosing marks are required for neutral and cationic ligand names, for names of inorganic anionic ligands containing multiplicative prefixes (such as triphosphato), for compositional names (such as carbon disulfide), for names of substituted organic ligands (even if there is no ambiguity in their use), and wherever necessary to avoid ambiguity. However, common ligand names such as aqua, ammine, carbonyl, nitrosyl, methyl, ethyl, *etc.*, do not require enclosing marks, unless there is ambiguity when they are absent.
- (iv) Ligands binding to metals through carbon atoms are treated in Chapter IR-10 on organometallic compounds.

Examples:

<i>Formula</i>	<i>Ligand name</i>
1. Cl^-	chlorido
2. CN^-	cyanido
3. H^-	hydrido ³
4. D^- or $^2\text{H}^-$	deuterido ³ or [² H]hydrido ³
5. $\text{PhCH}_2\text{CH}_2\text{Se}^-$	2-phenylethane-1-selenolato
6. MeCOO^-	acetato or ethanoato
7. Me_2As^-	dimethylarsanido
8. MeCONH_2	acetamide (<i>not</i> acetamido)
9. MeCONH^-	acetylazanido or acetylamido (<i>not</i> acetamido)
10. MeNH_2	methanamine

11. MeNH^-	methylazanido, or methylamido, or methanaminido (<i>cf.</i> Example 3 of Section IR-6.4.6)
12. MePH_2	methylphosphane
13. MePH^-	methylphosphanido
14. MeOS(O)OH	methyl hydrogen sulfite
15. MeOS(O)O^-	methyl sulfito, or methanolatodioxidosulfato(1-)

IR-9.2.2.4 *Charge numbers, oxidation numbers and ionic proportions*

The following methods can be used to assist in describing the composition of a compound:

- (i) The oxidation number of the central atom in a coordination entity may be indicated by a Roman numeral appended in parentheses to the central atom name (including the ending 'ate', if applicable), but only if the oxidation state can be defined without ambiguity. When necessary a negative sign is placed before the number. Arabic zero indicates the oxidation number zero.
- (ii) Alternatively, the charge on a coordination entity may be indicated. The net charge is written in arabic numbers, with the number preceding the charge sign, and enclosed in parentheses. It follows the name of the central atom (including the ending 'ate', if applicable) without the intervention of a space.
- (iii) The proportions of ionic entities in a coordination compound may be given by using multiplicative prefixes. (See Section IR-5.4.2.1.)

Examples:

1. $\text{K}_4[\text{Fe}(\text{CN})_6]$
potassium hexacyanidoferrate(II), or
potassium hexacyanidoferrate(4-), or
tetrapotassium hexacyanidoferrate
2. $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
hexaamminecobalt(III) chloride
3. $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$
pentaamminechloridocobalt(2+) chloride
4. $[\text{CoCl}(\text{NH}_3)_4(\text{NO}_2)]\text{Cl}$
tetraamminechloridonitrito- κN -cobalt(III) chloride
5. $[\text{PtCl}(\text{NH}_2\text{Me})(\text{NH}_3)_2]\text{Cl}$
diamminechlorido(methanamine)platinum(II) chloride
6. $[\text{CuCl}_2\{\text{O}=\text{C}(\text{NH}_2)_2\}_2]$
dichloridobis(urea)copper(II)
7. $\text{K}_2[\text{PdCl}_4]$
potassium tetrachloridopalladate(II)
8. $\text{K}_2[\text{OsCl}_5\text{N}]$
potassium pentachloridonitridoosmate(2-)

9. $\text{Na}[\text{PtBrCl}(\text{NH}_3)(\text{NO}_2)]$
sodium amminebromidochloridonitrito- κN -platinate(1-)
10. $[\text{Fe}(\text{CNMe})_6]\text{Br}_2$
hexakis(methyl isocyanide)iron(II) bromide
11. $[\text{Co}(\text{en})_3]\text{Cl}_3$
tris(ethane-1,2-diamine)cobalt(III) trichloride

IR-9.2.3 **Formulae of coordination compounds**

A (line) formula of a compound is used to provide basic information about the constitution of the compound in a concise and convenient manner. Different applications may require flexibility in the writing of formulae. Thus, on occasion it may be desirable to violate the following guidelines in order to provide more information about the structure of the compound that the formula represents. In particular, this is the case for dinuclear compounds where a great deal of structural information can be provided by relaxing the ordering principles outlined in Section IR-9.2.3.1. (See also Section IR-9.2.5, particularly Section IR-9.2.5.5.)

IR-9.2.3.1 *Sequence of symbols within the coordination formula*

- (i) The central atom symbol(s) is (are) listed first.
- (ii) The ligand symbols (line formulae, abbreviations or acronyms) are then listed in alphabetical order (see Section IR-4.4.2.2).⁵ Thus, CH_3CN , MeCN and NCMe would be ordered under C, M and N respectively, and CO precedes Cl because single letter symbols precede two letter symbols. The placement of the ligand in the list does not depend on the charge of the ligand.
- (iii) More information is conveyed by formulae that show ligands with the donor atom nearest the central atom; this procedure is recommended wherever possible, even for coordinated water.

IR-9.2.3.2 *Use of enclosing marks*

The formula for the entire coordination entity, whether charged or not, is enclosed in square brackets. When ligands are polyatomic, their formulae are enclosed in parentheses. Ligand abbreviations are also usually enclosed in parentheses. The nesting order of enclosing marks is as given in Sections IR-2.2 and IR-4.2.3. Square brackets are used only to enclose coordination entities, and parentheses and braces are nested alternately.

Examples 1–11 in Section IR-9.2.2.4 illustrate the use of enclosing marks in formulae. Note also that in those examples there is no space between representations of ionic species within a formula.

IR-9.2.3.3 *Ionic charges and oxidation numbers*

If the formula of a charged coordination entity is to be written without that of any counterion, the charge is indicated outside the square bracket as a right superscript, with the number before the sign. The oxidation number of a central atom may be represented by a Roman numeral, which should be placed as a right superscript on the element symbol.

Examples:

1. $[\text{PtCl}_6]^{2-}$
2. $[\text{Cr}(\text{OH}_2)_6]^{3+}$
3. $[\text{Cr}^{\text{III}}(\text{NCS})_4(\text{NH}_3)_2]^-$
4. $[\text{Cr}^{\text{III}}\text{Cl}_3(\text{OH}_2)_3]$
5. $[\text{Fe}^{\text{II}}(\text{CO})_4]^{2-}$

IR-9.2.3.4 *Use of abbreviations*

Abbreviations can be used to represent complicated organic ligands in formulae (although they should not normally be used in names). When used in formulae they are usually enclosed in parentheses.

Guidelines for the formulation of ligand abbreviations are given in Section IR-4.4.4; examples of such abbreviations are listed alphabetically in Table VII with diagrams of most shown in Table VIII.

In cases where coordination occurs through one of several possible donor atoms of a ligand, an indication of that donor atom may be desirable. This may be achieved in names through use of the kappa convention (see Section IR-9.2.4.2) in which the Greek lower case kappa (κ) is used to indicate the donor atom. To some extent, this device may also be used in formulae. For example, if the glycinate anion (gly) coordinates only through the nitrogen atom, the abbreviation of the ligand would be shown as gly- κN , as in the complex $[\text{M}(\text{gly}-\kappa\text{N})_3\text{X}_3]$.

IR-9.2.4 **Specifying donor atoms**

IR-9.2.4.1 *General*

There is no need to specify the donor atom of a ligand that has only one atom able to form a bond with a central atom. However, ambiguity may arise when there is more than one possible donor atom in a ligand. It is then necessary to specify which donor atom(s) of the ligand is (are) bound to the central atom. This includes cases where a ligand can be thought of as being formed by removal of H^+ from a particular site in a molecule or ion. For example, acetylacetonate, MeCOCHCOMe^- , has the systematic ligand name 2,4-dioxopentane-3-ido, which does not, however, imply bonding to the central atom from the central carbon atom in the ligand. The donor atom can be specified as shown in IR-9.2.4.2.

The only cases where specification of the donor atom is not required for a ligand that can bind to a central atom in more than one way are:

- monodentate O-bound carboxylate groups
- monodentate C-bound cyanide (ligand name 'cyanido')
- monodentate C-bound carbon monoxide (ligand name 'carbonyl')
- monodentate N-bound nitrogen monoxide (ligand name 'nitrosyl').

By convention, in these cases the ligand names imply the binding mode shown.

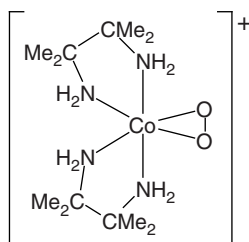
The following sections detail the means by which donor atoms are specified. The kappa (κ) convention, introduced in Section IR-9.2.4.2, is general and can be used for systems of great complexity. In some cases it may be simplified to the use of just the donor atom symbol (see Section IR-9.2.4.4).

These systems may be used in names, but they are not always suitable for use in formulae. The use of donor atom symbols is possible in the formulae of simple systems (see Section IR-9.2.3.4), but care must be taken to avoid ambiguity. The kappa convention is not generally compatible with the use of ligand abbreviations.

These methods are normally used only for specifying bonding between the central atom and isolated donor atoms. The eta (η) convention is used for any cases where the central atom is bonded to contiguous donor atoms within one ligand (see IR-10.2.5.1). Most examples of this latter kind are organometallic compounds (Chapter IR-10) but the example below shows its use for a coordination compound.

Example:

1.



bis(2,3-dimethylbutane-2,3-diamine)(η^2 -peroxido)cobalt(1+)

IR-9.2.4.2 The kappa convention

Single ligating atoms are indicated by the italicized element symbol preceded by a Greek kappa, κ . These symbols are placed after the portion of the ligand name that represents the ring, chain or substituent group in which the ligating atom is found.

Example:

1. $[\text{NiBr}_2(\text{Me}_2\text{PCH}_2\text{CH}_2\text{PMe}_2)]$

dibromido[ethane-1,2-diylbis(dimethylphosphane- κP)]nickel(II)

Multiplicative prefixes which apply to a ligand or portions of a ligand also apply to the donor atom symbols. In some cases this may require the use of an alternative ligand name, *e.g.* where multiplicative prefixes can no longer be used because the ligation of otherwise equivalent portions of the ligand is different. Several examples of this are given below.

Simple examples are thiocyanato- κN for nitrogen-bonded NCS and thiocyanato- κS for sulfur-bonded NCS. Nitrogen-bonded nitrite is named nitrito- κN and oxygen-bonded nitrite is named nitrito- κO , as in pentaamminenitrito- κO -cobalt(III).

For ligands with several ligating atoms linearly arranged along a chain, the order of κ symbols should be successive, starting at one end. The choice of end is based upon alphabetical order if the ligating atoms are different, *e.g.* cysteinato- $\kappa N, \kappa S$; cysteinato- $\kappa N, \kappa O$.

Donor atoms of a particular element may be distinguished by adding a right superscript numerical locant to the italicized element symbol or, in simple cases (such as Example 3 below), a prime or primes.

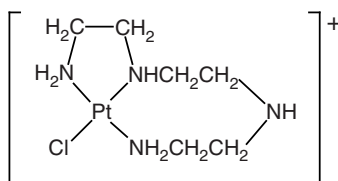
Superscript numerals, on the other hand, are based on an appropriate numbering of some or all of the atoms of the ligand, such as numbering of the skeletal atoms in parent hydrides, and allow the position of the bond(s) to the central atom to be specified even in quite complex cases. In the simple case of acetylacetonate, MeCOCHCOMe^- , mentioned above, the ligand name 2,4-dioxopentan-3-ido- κ^3 would imply ligation by the central carbon atom in the pentane skeleton (see also Example 4 below).

In some cases, standard nomenclature procedures do not provide locants for the donor atoms in question. In such cases simple *ad hoc* procedures may be applicable. For example, for the ligand $(\text{CF}_3\text{COCHCOMe})^-$, the name 1,1,1-trifluoro-2,4-dioxopentan-3-ido- κO could be used to refer to coordination, through oxygen, of the CF_3CO portion of the molecule, while coordination by MeCO would be identified by 1,1,1-trifluoro-2,4-dioxopentan-3-ido- $\kappa O'$. The prime indicates that the MeCO oxygen atom is associated with a higher locant in the molecule than the CF_3CO oxygen atom. The oxygen atom of the CF_3CO portion of the ligand is attached to C2, while that of MeCO is attached to C4. Alternatively, the name could be modified to 1,1,1-trifluoro-2-(oxo- κO)-4-oxopentan-3-ido and 1,1,1-trifluoro-2-oxo-4-(oxo- κO)pentan-3-ido, respectively, for the two binding modes above.

In cases where two or more identical ligands (or parts of a polydentate ligand) are involved, a superscript is used on κ to indicate the number of such ligations. As mentioned above, any multiplicative prefixes for complex entities are presumed to operate on the κ symbol as well. Thus, one uses the partial name ‘...bis(2-amino- κN -ethyl)...’ and not ‘...bis(2-amino- $\kappa^2 N$ -ethyl)...’ in Example 2 below. Examples 2 and 3 use tridentate chelation by the linear tetraamine ligand *N,N'*-bis(2-aminoethyl)ethane-1,2-diamine to illustrate these rules.

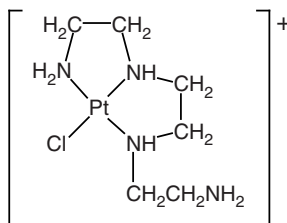
Examples:

2.



[*N,N'*-bis(2-amino- κN -ethyl)ethane-1,2-diamine- κN]chloridoplatinum(II)

3.



[*N*-(2-amino- κN -ethyl)-*N'*-(2-aminoethyl)ethane-1,2-diamine- $\kappa^2 N,N'$] chloridoplatinum(II)

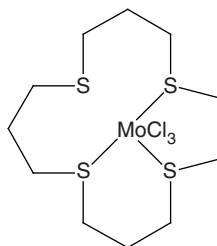
Example 2 illustrates how coordination by the two terminal primary amino groups of the ligand is indicated by placing the kappa index after the substituent group name and within the effect of the ‘bis’ doubling prefix. The appearance of the simple index κN after the ‘ethane-1,2-diamine’ indicates the binding by only one of the two equivalent secondary amino nitrogen atoms.

Only one of the primary amines is coordinated in Example 3. This is indicated by not using the doubling prefix ‘bis’, repeating (2-aminoethyl), and inserting the κ index only in the first such unit, *i.e.* (2-amino- κN -ethyl). The involvement of both of the secondary ethane-1,2-diamine nitrogen atoms in chelation is indicated by the index $\kappa^2 N, N'$.

Tridentate chelation by the tetrafunctional macrocycle in Example 4 is shown by the kappa index following the ligand name. The ligand locants are required in order to distinguish this complex from those where the central atom is bound to other combinations of the four potential donor atoms.

Example:

4.

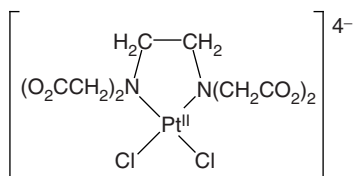


trichlorido(1,4,8,12-tetrathiacyclopentadecane- $\kappa^3 S^{1,4,8}$)molybdenum, or
trichlorido(1,4,8,12-tetrathiacyclopentadecane- $\kappa^3 S^1, S^4, S^8$)molybdenum

Well-established modes of chelation of the (ethane-1,2-diyl)dinitrilo)tetraacetato ligand (edta), namely bidentate, tetradentate and pentadentate, are illustrated in Examples 5–8. The multiplicative prefix ‘tetra’ used in Example 5 cannot be used in Examples 6 and 7 because of the need to avoid ambiguity about which acetate arms are coordinated to the central atom. In such cases the coordinated fragments are cited before the uncoordinated fragments in the ligand name. Alternatively, a modified name may be used, as in Example 7, where the use of the preferred IUPAC name *N,N'*-ethane-1,2-diylbis[*N*-(carboxymethyl)glycine] (see Section P-44.4 of Ref. 1) is demonstrated.

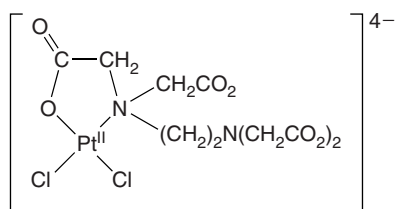
Examples:

5.

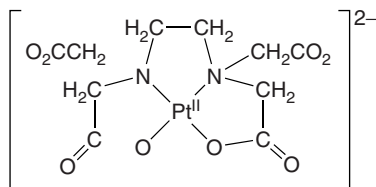


dichlorido[(ethane-1,2-diyl)dinitrilo- $\kappa^2 N, N'$]tetraacetatoplatinate(4-)

6.

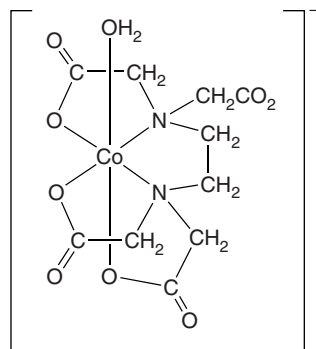
dichlorido[(ethane-1,2-diyl dinitrilo- κN)(acetato- κO)triacetato]platinate(II)

7.



[(ethane-1,2-diyl dinitrilo- $\kappa^2 N, N'$)(N, N' -diacetato- $\kappa^2 O, O'$)(N, N' -diacetato)]platinate(2-), or
 { N, N' -ethane-1,2-diylbis[N -(carboxylatomethyl)glycinato- $\kappa O, \kappa N$]}platinate(2-)

8.



aqua[(ethane-1,2-diyl dinitrilo- $\kappa^2 N, N'$)tris(acetato- κO)acetato]cobaltate(1-),
 or aqua[N -{2-[bis(carboxylato- κO -methyl)amino- κ]-ethyl}- N -(carboxylato- κO -methyl)glycinato- κ]-cobaltate(1-)]

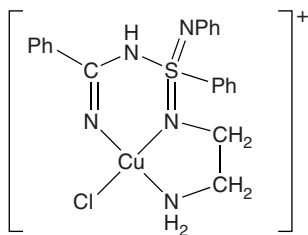
A compound of edta in which one amino group is not coordinated while all four carboxylato groups are bound to a single metal ion would bear the ligand name (ethane-1,2-diyl dinitrilo- κN)tetrakis(acetato- κO) within the name of the complex.

The mixed sulfur-oxygen cyclic polyether 1,7,13-trioxa-4,10,16-trithiacyclooctadecane might chelate to alkali metals only through its oxygen atoms and to second-row transition elements only through its sulfur atoms. The corresponding kappa indexes for such chelate complexes would be $\kappa^3 O^1, O^7, O^{13}$ and $\kappa^3 S^4, S^{10}, S^{16}$, respectively.

Examples 9–11 illustrate three modes of chelation of the ligand N -[N -(2-aminoethyl)- N' , S -diphenylsulfonodiimidoyl]benzenimidamide. The use of kappa indexes allows these binding modes (and others) to be distinguished and identified, in spite of the abundance of heteroatoms that could coordinate.

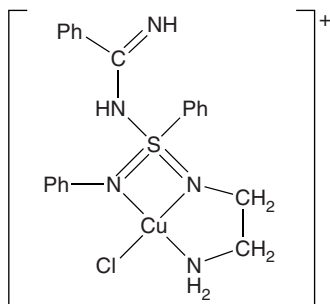
Examples:

9.



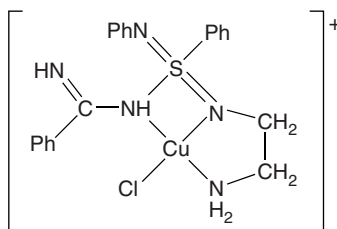
{*N*-[*N*-(2-amino-κ*N*-ethyl)-*N'*,*S*-diphenylsulfonodiimidoyl-κ*N*]benzenimidamide-κ*N'*}chloridocopper(II)

10.



{*N*-[*N*-(2-amino-κ²*N,N'*-ethyl)-*N'*,*S*-diphenylsulfonodiimidoyl-κ*N*]benzenimidamide}chloridocopper(II)

11.



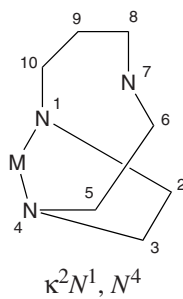
{*N*-[*N*-(2-amino-κ*N*-ethyl)-*N'*,*S*-diphenylsulfonodiimidoyl-κ*N*]benzenimidamide-κ*N*}chloridocopper(II)

The distinction between the names in Examples 9 and 11 rests on the conventional priming of the imino nitrogen atom in the benzenimidamide functional group. The prime differentiates the imino benzenimidamide nitrogen atom from that which is substituted (and unprimed at the beginning of the name).

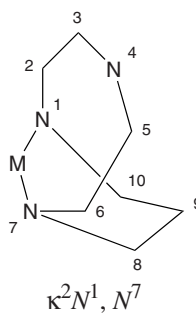
The use of donor atom locants on the atomic symbols to indicate point of ligation is again illustrated by the two isomeric bidentate modes of binding of the macrocycle 1,4,7-triazecane (or 1,4,7-triazacyclodecane) (Examples 12 and 13). Conveying the formation of the five-membered chelate ring requires the index κ²*N*¹,*N*⁴, while the six-membered chelate ring requires the index κ²*N*¹,*N*⁷. Example 14 shows that due to the local nature of the locants used with κ, the same locant and atomic symbol may appear several times, referring to different parts of the ligand.

Examples:

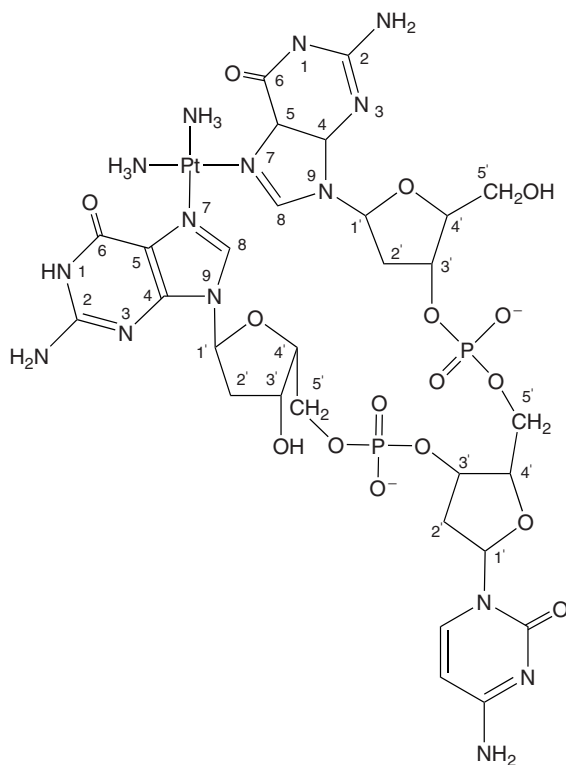
12.



13.



14.



diammine[2'-deoxyguanylyl- κN^7 -(3'→5')-2'-deoxycytidylyl(3'→5')-2'-deoxyguanosinato- κN^7 (2-)]platinum(II)

IR-9.2.4.3 *Comparison of the eta and kappa conventions*

The eta convention (Section IR-10.2.5.1) is applied in cases where contiguous donor atoms within a given ligand are involved in bonding to a central atom. Thus, it is used only when there is more than one ligating atom, and the term η^1 is not used. The contiguous atoms are often the same element, but need not be.

The kappa convention is used to specify bonding from isolated donor atoms to one or more central atoms.

In cases where two or more identical ligands (or parts of a polydentate ligand) are bound to a central atom, a superscript is used on κ to indicate the number of donor atom-to-central atom bonds.

IR-9.2.4.4 *Use of donor atom symbol alone in names*

In certain cases the kappa convention may be simplified. Donor atoms of a ligand may be denoted by adding only the italicized symbol(s) for the donor atom (or atoms) to the end of the name of the ligand. Thus, for the 1,2-dithiooxalate anion, ligand names such as 1,2-dithiooxalato- $\kappa S, \kappa S'$ and 1,2-dithiooxalato- $\kappa O, \kappa S$ may, with no possibility of confusion, be shortened to 1,2-dithiooxalato- S, S' and 1,2-dithiooxalato- O, S , respectively. Other examples are thiocyanato- N and thiocyanato- S , and nitrito- N and nitrito- O .

IR-9.2.5 **Polynuclear complexes**

IR-9.2.5.1 *General*

Polynuclear inorganic complexes exist in a bewildering array of structural types, such as ionic solids, molecular polymers, extended assemblies of oxoanions, chains and rings, bridged metal complexes, and homonuclear and heteronuclear clusters. This section primarily treats the nomenclature of bridged metal complexes and homonuclear and heteronuclear clusters. Coordination polymers are treated extensively elsewhere.⁶

As a general principle, as much structural information as possible should be presented when writing the formula or name of a polynuclear complex. However, polynuclear complexes may have structures so large and extended as to make a rational structure-based nomenclature impractical. Furthermore, their structures may be undefined or not suitably elucidated. In such cases, the principal function of the name or formula is to convey the stoichiometric proportions of the various moieties present.

In the present and following sections, particular complexes are often used as examples several times to show how they may be named differently according to whether only stoichiometry is to be specified or partial or complete structural information is to be included.

Ligands in polynuclear complexes are cited in alphabetical order both in formulae and names. The number of each ligand is specified by subscript numerical multipliers in formulae (Sections IR-9.2.3.1 to IR-9.2.3.4) and by appropriate multiplicative prefixes in names (Sections IR-9.2.2.1 to IR-9.2.2.3). The number of central atoms of a given kind, if greater than one, is indicated similarly.

Note, however, that the rules for formula writing may be relaxed in various ways in order better to display particular features of the structures in question. Use is made of this flexibility in many examples below.

Example:

1. $[\text{Rh}_3\text{H}_3\{\text{P}(\text{OMe})_3\}_6]$
trihydridoheptakis(trimethyl phosphite)trirhodium

If there is more than one element designated as a central atom, these elements are ranked according to the order in which they appear in Table VI. The later an element appears in the sequence of Table VI, the earlier it comes in the list of central atom symbols in the formula as well as in the list of central atom names in the name of the complex.

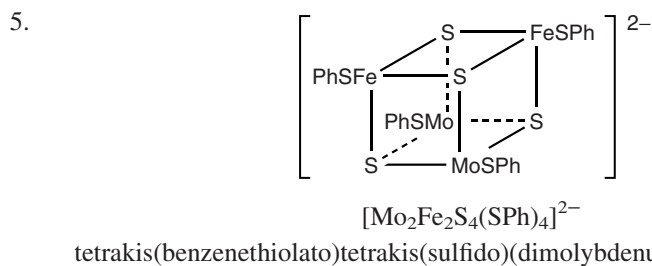
Example:

2. $[\text{ReCo}(\text{CO})_9]$ nonacarbonylrheniumcobalt

For anionic species, the ending 'ate' and the charge number (see Section IR-5.4.2.2) are added after the central atom list which is enclosed in parentheses if more than one element is involved.

Examples:

3. $[\text{Cr}_2\text{O}_7]^{2-}$ heptaoxidodichromate(2-)
4. $[\text{Re}_2\text{Br}_8]^{2-}$ octabromidodirhenate(2-)



Although not extensively exemplified here, it is worth noting that the formalism developed below for polynuclear complexes is applicable also to (formal) complexes in which the central atoms are not metals.

Example:

6. $[\text{PSO}_7]^{2-}$ heptaoxido(phosphorussulfur)ate(2-)

A number of oxoacids and related species are given such names in Chapter IR-8 and Table IX.

The symbol kappa, κ , was introduced, in Section IR-9.2.4.2, in order to specify the ligating atoms in polyatomic ligands. This use also applies to such ligands when they appear in polynuclear complexes. However, the symbol κ then assumes a new function, namely that

of specifying which ligating atoms bind to which central atom. In order to do this, the central atoms must be identified, *i.e.* by assigning numbers to these atoms according to the order in which they appear in the central atom list. (The later the central atom elements appear in Table VI, the lower the numbers they are assigned.)

Additional rules are needed when there is more than one central atom of the same element (see Sections IR-9.2.5.5 and IR-9.2.5.6) except if the presence of symmetry in the structure makes two or more of the central atoms equivalent (see, for example, Section IR-9.2.5.4) and the name eventually generated is independent of the numbering.

The central atom numbers are then used as locants for the ligating atoms and are placed to the left of each kappa symbol. Individual kappa designators, *i.e.* kappa symbols with a numerical superscript (as applicable), central atom locant and ligator atom symbol, are separated by commas.

Examples:

7. $[(OC)_5Re^1Co^2(CO)_4]$
nonacarbonyl-1 κ^5C ,2 κ^4C -rheniumcobalt
8. $[Cl_4Re^1Re^2Cl_4]^{2-}$
octachlorido-1 κ^4Cl ,2 κ^4Cl -dirhenate(2-)

In these two examples, structural information indicated by the formulae is not communicated by the names. In fact, any polynuclear complex must either contain at least one ligand binding to more than one central atom (a bridging ligand) or contain a bond between two central atoms. In order to specify these aspects of the structure in names, further devices are needed. These are introduced in the following two sections.

IR-9.2.5.2 *Bridging ligands*

Bridging ligands, as far as they can be specified, are indicated by the Greek letter μ appearing before the ligand symbol or name and separated from it by a hyphen; the conventions applied were briefly introduced in IR-9.1.2.10. In names, the whole term, *e.g.* μ -chlorido, is separated from the rest of the name by hyphens, as in ammine- μ -chlorido-chlorido, *etc.*, unless the bridging ligand name is contained within its own set of enclosing marks. If the bridging ligand occurs more than once, multiplicative prefixes are employed, as in tri- μ -chlorido-chlorido, or as in bis(μ -diphenylphosphanido), if more complex ligand names are involved.

Bridging ligands are listed in alphabetical order together with the other ligands, but in names a bridging ligand is cited before a corresponding non-bridging ligand, as in di- μ -chlorido-tetrachlorido. In formulae, bridging ligands are placed after terminal ligands of the same kind. Thus, in both names and formulae bridging ligands are placed further away from the central atoms than are terminal ligands of the same kind.

Example:

1. $[Cr_2O_6(\mu-O)]^{2-}$ μ -oxido-hexaoxidodichromate(2-)

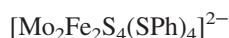
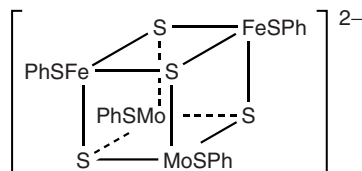
The bridging index n , the number of coordination centres connected by a bridging ligand, is placed as a right subscript. The bridging index 2 is not normally indicated. Multiple

bridging is listed in descending order of complexity, *e.g.* μ_3 -oxido-di- μ -oxido-trioxido. For ligand names requiring enclosing marks, μ is contained within those marks.

The kappa convention is used together with μ when it is necessary to specify which central atoms are bridged, and through which donor atoms. The kappa descriptor counts all donor atom-to-central atom bonds so that in Example 2 below the descriptor $1:2:3\kappa^3S$ specifies all three bonds from the sulfur atom bridging central atoms 1, 2 and 3.

Example:

2.



tetrakis(benzenethiolato)- $1\kappa S, 2\kappa S, 3\kappa S, 4\kappa S$ -tetra- μ_3 -sulfido-
 $1:2:3\kappa^3S; 1:2:4\kappa^3S; 1:3:4\kappa^3S; 2:3:4\kappa^3S$ -(dimolybdenumdiiron)ate(2-)

Here, the two molybdenum atoms are numbered 1 and 2 and the two iron atoms 3 and 4 according to the rule in Section IR-9.2.5.1. Due to the symmetry of the compound, it is not necessary to distinguish between 1 and 2 or between 3 and 4.

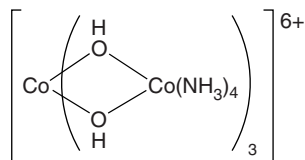
Example:



When single ligating atoms bind to two or more central atoms, the central atom locants are separated by a colon. For example, tri- μ -chlorido- $1:2\kappa^2Cl; 1:3\kappa^2Cl; 2:3\kappa^2Cl$ - indicates that there are three bridging chloride ligands and they bridge between central atoms 1 and 2, 1 and 3, and 2 and 3. Note that because of the use of the colon, sets of bridge locants are separated here by semicolons rather than commas.

Example:

4.

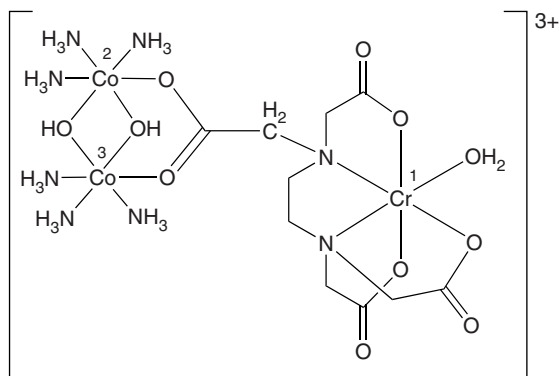


dodecaammine- $1\kappa^4N, 2\kappa^4N, 3\kappa^4N$ -hexa- μ -hydroxido-
 $1:4\kappa^4O; 2:4\kappa^4O; 3:4\kappa^4O$ -tetracobalt(6+)

The central atom locants given in this example are assigned by following the rules in Sections IR-9.2.5.5 and IR-9.2.5.6. In this case, the central cobalt atom is assigned the locant 4.

Example:

5.



hexaammine-2 κ^3N ,3 κ^3N -aqua-1 κO -{ μ_3 -(ethane-1,2-diyl dinitrilo-1 κ^2N,N')-tetraacetato-1 κ^3O^1,O^2,O^3 :2 κO^4 :3 κO^4 }-di- μ -hydroxido-2:3 κ^4O -chromiumdicobalt(3+)

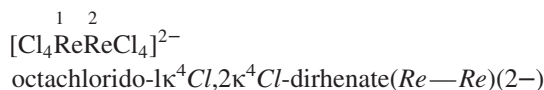
In this name, the obvious numbering (1,1',2,2',3,3',4,4') of the oxygen ligating atoms of the four carboxylate groups is tacitly assumed.

IR-9.2.5.3 Metal-metal bonding

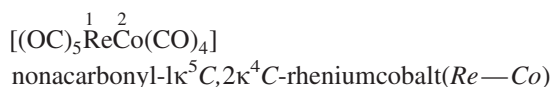
Metal-metal bonding or, more generally, bonding between central atoms in complexes, may be indicated in names by placing italicized atomic symbols of the appropriate central atoms, separated by an 'em' dash and enclosed in parentheses, after the list of central atom names and before the ionic charge. The central atom element symbols are placed in the same order as the central atoms appear in the name (*i.e.* according to Table VI, with the first element reached when following the arrow being placed last). The number of such bonds is indicated by an arabic numeral placed before the first element symbol and separated from it by a space. For the purpose of nomenclature, no distinction is made between different bond orders. If there is more than one central atom of an element present in the structure, and it is necessary to indicate which of them is involved in the bond in question (because they are inequivalent), the central atom locant (see Section IR 9.2.5.6) can be placed as a superscript immediately after the element symbol, as shown in Example 4.

Examples:

1.



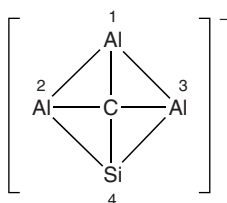
2.



3.



4.



μ_4 -carbido-*quadro*-
(trialuminumsilicon)ate (Al^1-Al^2) (Al^1-Al^3) (Al^2-Si) (Al^3-Si) (1-)

(Examples 3 and 4 include the structural descriptors *triangulo* and *quadro* which are introduced below in Section IR-9.2.5.7.) Note that the name in Example 3 does not specify which chloride ligands bind to which central atoms.

IR-9.2.5.4 Symmetrical dinuclear entities

For symmetrical dinuclear entities, the name may be simplified by employing multiplicative prefixes.

Examples:

1. $[Re_2Br_8]^{2-}$
bis(tetrabromidorhenate)(*Re—Re*)(2-)
2. $[Mn_2(CO)_{10}]$
bis(pentacarbonylmanganese)(*Mn—Mn*)
3. $[{Cr(NH_3)_5}_2(\mu-OH)]^{5+}$
 μ -hydroxido-bis(pentaamminechromium)(5+)
4. $[{PtCl(PPh_3)}_2(\mu-Cl)_2]$
di- μ -chlorido-bis[chlorido(triphenylphosphane)platinum]
5. $[{Fe(NO)_2}_2(\mu-PPh_2)_2]$
bis(μ -diphenylphosphanido)bis(dinitrosyliron)
6. $[{Cu(py)}_2(\mu-O_2CMe)_4]$
tetrakis(μ -acetato- $\kappa O:\kappa O'$)bis[(pyridine)copper(II)]

In some cases multiplicative prefixes may also be used to simplify names of unsymmetrical complexes (see Example 5 in Section IR-9.2.5.5).

IR-9.2.5.5 Unsymmetrical dinuclear entities

The name of an unsymmetrical dinuclear species will result from following the general rules described in Sections IR-9.2.5.1 to IR-9.2.5.3.

Example:

1. $[ClHgIr(CO)Cl_2(PPh_3)_2]$
carbonyl-1 κC -trichlorido-1 $\kappa^2 Cl$, 2 κCl -bis(triphenylphosphane-1 κP)iridiummercury(*Ir—Hg*)

In this example, iridium is reached last on following the arrow shown in Table VI. It is therefore listed before mercury in the name and is given the central atom locant 1.

The only remaining problem is to number the central atoms in cases where they are the same but have different coordination environments. In this case, the central atom with the larger coordination number is given the lower number (locant), if applicable. If the coordination numbers are equal, the central atom with the greater number of ligands or ligating atoms represented earlier in the name is given the lower number (locant). Thus, in Example 2 the chromium atom with five of the nine ammine ligands attached is given priority number 1.

Examples:

2. $[(\text{H}_3\text{N})_5\text{Cr}(\mu\text{-OH})\text{Cr}(\text{NH}_2\text{Me})(\text{NH}_3)_4]^{5+}$
nonaammine-1 $\kappa^5\text{N}$,2 $\kappa^4\text{N}$ - μ -hydroxido-(methanamine-2 κN)dichromium(5+)
3. $[(\text{H}_3\text{N})_3\text{Co}(\mu\text{-NO}_2)(\mu\text{-OH})_2\text{Co}(\text{NH}_3)_2(\text{py})]^{3+}$
pentaammine-1 $\kappa^3\text{N}$,2 $\kappa^2\text{N}$ -di- μ -hydroxido- μ -nitrito-1 κN :2 κO -
(pyridine-2 κN)dicobalt(3+)
4. $[(\text{bpy})(\text{H}_2\text{O})\text{Cu}(\mu\text{-OH})_2\text{Cu}(\text{bpy})(\text{SO}_4)]$
aqua-1 κO -(2,2'-bipyridine-1 $\kappa^2\text{N}$, N' ')(2,2'-bipyridine-2 $\kappa^2\text{N}$, N')-di- μ -hydroxido-
(sulfato-2 κO)dicopper(II)

In some cases, it is not necessary to number explicitly the two differently coordinated central atoms to arrive at a name, as shown in Example 5. Note the use of a multiplicative prefix to simplify the name, as also demonstrated in Section IR-9.2.5.4 for fully symmetrical structures.

Example:

5. $[\{\text{Co}(\text{NH}_3)_3\}_2(\mu\text{-NO}_2)(\mu\text{-OH})_2]^{3+}$
di- μ -hydroxido- μ -nitrito- κN : κO -bis(triamminecobalt)(3+)

IR-9.2.5.6 Trinuclear and larger structures

The methods described in the preceding sections for naming ligands and designating ligating atoms are general, and applicable irrespective of the nuclearity (the number of central atoms involved). However, in most cases numbering of the central atoms is needed in order to construct a systematic additive name for a coordination entity. Obtaining such a numbering is the part of the naming process which becomes increasingly complex in the general case as the nuclearity increases. This section suggests general procedures for assigning locant numbers to central atoms.

If no two central atoms are the same element, locant numbers for the central atoms and the order they appear in the name can be determined using Table VI. The first central atom reached on following the arrow in the Table receives the highest locant number, while the last reached is given the locant 1. This method can also be applied to systems where there is more than one of a given type of central atom, provided there is symmetry present in the structure that makes all of the central atoms of a given element equivalent. Indeed, in the extreme case, it may not be necessary to assign locants at all, provided all the central atoms are equivalent.

Examples:

1. $[\text{Be}_4(\mu_4\text{-O})(\mu\text{-O}_2\text{CMe})_6]$
hexakis(μ -acetato- $\kappa\text{O}:\kappa\text{O}'$)- μ_4 -oxido-*tetrahedro*-tetraberyllium
2. $[\text{Os}_3(\text{CO})_{12}]$
dodecacarbonyl-1 $\kappa^4\text{C}$,2 $\kappa^4\text{C}$,3 $\kappa^4\text{C}$ -*triangulo*-triosmium(3 Os—Os)

(The descriptors *tetrahedro* and *triangulo* are introduced in Section IR-9.2.5.7.)

Another such case is Example 5 in Section IR-9.2.5.2 where it is immaterial which of the two cobalt atoms is given number 2 and which one number 3. The systematic name will be the same.

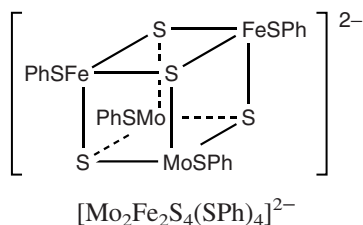
The proposed *general procedure* for constructing a coordination-type additive name for a polynuclear entity is as follows:

- (i) Identify the central atoms and ligands.
- (ii) Name the ligands, including κ , η and μ designators (except for the central atom locants). Note that ligand names may have to be modified if κ , η or μ symbols apply only to some portions of the ligand that are otherwise equivalent (and described by a multiplicative prefix such as 'tri' or 'tris').
- (iii) Place ligand names in alphabetical order.
- (iv) Assign central atom locants by applying the following rules:
 - (a) Apply the element sequence of Table VI. The later an element is met when following the arrows, the lower its locant number. This criterion will determine the numbering if all central atoms are different elements. Locants may be assigned to atoms of the same element by applying the next rules.
 - (b) Within each class of identical central atoms, assign lower locant numbers to central atoms with higher coordination numbers.
 - (c) Proceed through the alphabetical list of ligand names. Examine the names or name parts specifying ligating atoms explicitly (as in a κ or η designator) or implicitly (as in the ligand name 'carbonyl'). As soon as a subset of ligating atoms is met which is not evenly distributed among the central atoms still awaiting the assignment of distinct locant numbers, the central atoms with the most ligating atoms of this kind are given the lowest numbers available. This process of sequential examination of the ligands is continued until all central atoms have been assigned locants or all ligands have been considered.
 - (d) Any central atoms that are inequivalent and have not yet been assigned distinct locant numbers will differ only in the other central atoms to which they are directly bonded. The locant numbers of these directly bonded neighbouring central atoms are compared and the central atom with the lowest-locant neighbouring atoms is given the lowest of the remaining possible locants (see Example 9 below).

Note that the central atom locants assigned using these rules need not coincide with those assigned when using other types of nomenclature such as substitutive nomenclature (*cf.* Chapter IR-6), if that is applicable, or the nomenclature systems described in Chapters II-1 or II-5 of Ref. 7.

Example:

3.

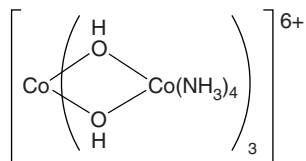


tetrakis(benzenethiolato)-1 κ S,2 κ S,3 κ S,4 κ S-tetra- μ_3 -sulfido-
1:2:3 κ^3 S;1:2:4 κ^3 S;1:3:4 κ^3 S;2:3:4 κ^3 S-(dimolybdenumdiiron)ate(2-)

Using the rules above, no distinction is obtained between the two molybdenum atoms or between the two iron atoms. However, no distinction is needed.

Example:

4.

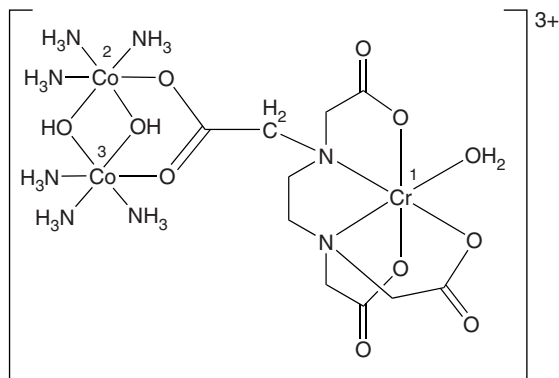


dodecaammine-1 κ^4 N,2 κ^4 N,3 κ^4 N-hexa- μ -hydroxido-
1:4 κ^4 O;2:4 κ^4 O;3:4 κ^4 O-tetracobalt(6+)

Rules (a) and (b) do not result in a distinction between the four cobalt atoms. By rule (c), however, the three peripheral cobalt atoms are assigned numbers 1, 2 and 3 because they carry the ammine ligands appearing first in the name, and the central cobalt atom is thus number 4. This is all that is required to construct the name, because of the symmetry of the complex.

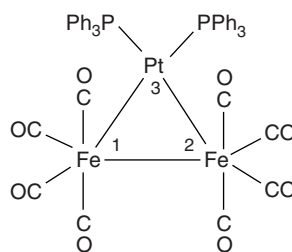
Examples:

5.

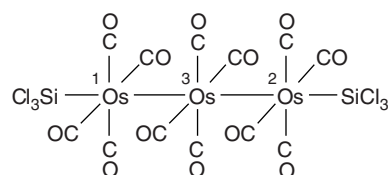


hexaammine-2 κ^3 N,3 κ^3 N-aqua-1 κ O-{ μ_3 -(ethane-1,2-diyl dinitrilo-1 κ^2 N,N')-
tetraacetato-1 κ^3 O¹,O²,O³:2 κ O⁴:3 κ O^{4'}}-di- μ -hydroxido-2:3 κ^4 O-
chromiumdicobalt(3+)

6.



octacarbonyl-1 κ^4 C,2 κ^4 C-bis(triphenylphosphane-3 κ P)-triangulo-diironplatinum(*Fe—Fe*)(2 *Fe—Pt*)

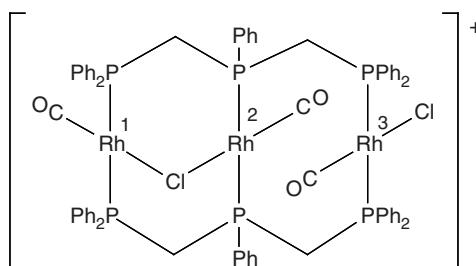
7. [Os₃(CO)₁₂(SiCl₃)₂]

dodecacarbonyl-1 κ^4 C,2 κ^4 C,3 κ^4 C-bis(trichlorosilyl)-1 κ Si,2 κ Si-triosmium(*Os*¹—*Os*³)(*Os*²—*Os*³)

All three osmium atoms have four carbonyl ligands. The two osmium atoms with trichlorosilyl ligands are assigned central atom locants 1 and 2, as these ligands are the first that are not evenly distributed. Symmetry in the structure means that the locants 1 and 2 can be assigned either way around.

Example:

8.



tricarboxyl-1 κ C,2 κ C,3 κ C- μ -chlorido-1:2 κ^2 Cl-chlorido-3 κ Cl-bis{ μ_3 -bis[(diphenylphosphanyl)methyl]-1 κ P:3 κ P'-phenylphosphane-2 κ P}trirhodium(1+)

or, using the preferred IUPAC name¹ for the phosphane ligand:

tricarboxyl-1 κ C,2 κ C,3 κ C- μ -chlorido-1:2 κ^2 Cl-chlorido-3 κ Cl-bis{ μ_3 -[phenylphosphanediy]-1 κ P-bis(methylene)]bis(diphenylphosphane)-2 κ P':3 κ P''}trirhodium(1+)

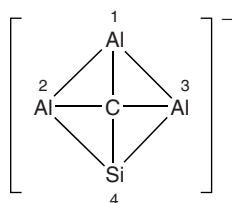
Example 8 illustrates how using different (equally systematic) names for ligands may result in different additive names and different locant numberings.

In the first name, the first place where the rhodium atoms can be identified as being inequivalent is at the kappa term associated with the μ -chlorido ligand. Thus, the chloride-bridged rhodium atoms must be assigned the central atom locants 1 and 2 (although which is which is not known at this stage), and the other rhodium atom must be assigned the locant 3. The next difference in the name that relates to central atom 1 or 2 is the diphenylphosphanyl κ term. Those portions of the ligand are bound to the end rhodium atoms and not to the middle rhodium atom. Since one of the end rhodium atoms is already given the locant 3, from the earlier difference, the other rhodium atom must be assigned locant 1, and the middle atom is left with locant 2.

For the second name, the locant 3 is assigned in the same way, but the middle Rh atom should be assigned locant 1 as it now appears earlier in the ligand name (in the κ term for phosphanediy).

Example:

9.



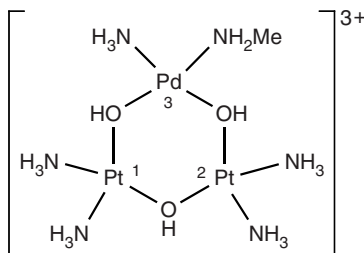
μ_4 -carbido-*quadro*-
(trialuminumsilicon)ate(Al^1-Al^2)(Al^1-Al^3)(Al^2-Si)(Al^3-Si)(1-)

In this example the central atom locants are assigned as follows. Rule (a), above, results in the silicon atom being assigned locant 4. The coordination numbers and ligand distribution are the same for the three aluminium atoms, which only differ in which other central atoms they are bonded to. The numbering of the aluminium atoms follows from rule (d) above.

The prefix '*cyclo*', italicized and cited before all ligands, may be used for monocyclic compounds.

Example:

10.

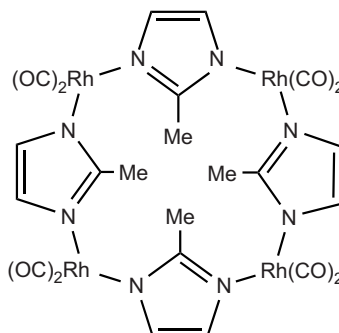


cyclo-pentaammine-1 κ^2N ,2 κ^2N ,3 κN -tri- μ -hydroxido-
1:2 κ^2O ;1:3 κ^2O ;2:3 κ^2O -(methanamine-3 κN)diplatinumpalladium(3+)

The two platinum atoms are equivalent and receive lower central atom locants than palladium by rule (a).

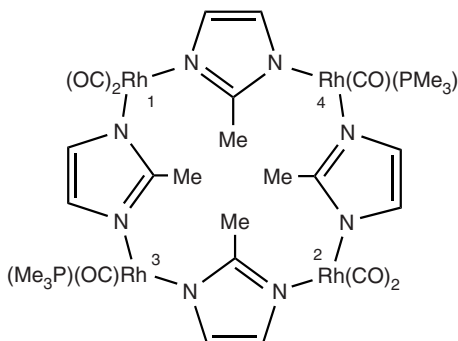
Examples:

11.



cyclo-tetrakis(μ-2-methylimidazolido-κN¹:κN³)tetrakis(dicarbonylrhodium)

12.



cyclo-hexacarbonyl-1κ²C,2κ²C,3κC,4κC-tetrakis(μ-2-methyl-1H-imidazol-1-ido)-1:3κ²N¹:N³;1:4κ²N³:N¹;2:3κ²N³:N¹;2:4κ²N¹:N³-bis(trimethylphosphane)-3κP,4κP-tetrrhodium

IR-9.2.5.7 Polynuclear clusters: symmetrical central structural units

The structural features of complex polynuclear entities may be communicated using the concept of a *central structural unit* (CSU). Only the metal atoms are considered for this purpose. For nonlinear clusters, descriptors such as *triangulo*, *tetrahedro* and *dodecahedro* are used to describe central structural units in simple cases, as has already been exemplified above. However, synthetic chemistry has advanced far beyond the limited range of central structural units associated with this usage. A more comprehensive CSU descriptor and a numbering system, the CEP (Casey, Evans, Powell) system, has been developed specifically for fully triangulated polyboron polyhedra (deltahedra).⁸ These CEP descriptors may be used in general as systematic alternatives to the traditional descriptors for fully triangulated polyhedra (deltahedra). Examples are listed in Table IR-9.1.

Table IR-9.1 Structural descriptors

<i>Number of atoms in CSU</i>	<i>Descriptor</i>	<i>Point group</i>	<i>CEP descriptor</i>
3	<i>triangulo</i>	D_{3h}	
4	<i>quadro</i>	D_{4h}	
4	<i>tetrahedro</i>	T_d	$[T_d-(13)-\Delta^4-closo]$
5		D_{3h}	$[D_{3h}-(131)-\Delta^6-closo]$
6	<i>octahedro</i>	O_h	$[O_h-(141)-\Delta^8-closo]$
6	<i>triprismo</i>	D_{3h}	
8	<i>antiprismo</i>	S_6	
8	<i>dodecahedro</i>	D_{2d}	$[D_{2d}-(2222)-\Delta^6-closo]$
8	<i>hexahedro (cube)</i>	O_h	
12	<i>icosahedro</i>	I_h	$[I_h-(1551)-\Delta^{20}-closo]$

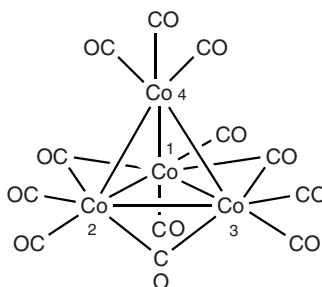
In brief, the numbering of the CSU is based on locating a reference axis and planes of atoms perpendicular to the reference axis. The reference axis is the axis of highest rotational symmetry. Select that end of the reference axis with a single atom (or smallest number of atoms) in the first plane to be numbered. Orient the CSU so that the first position to receive a locant in the first plane with more than one atom is in the twelve o'clock position. Assign locant numbers to the axial position or to each position in the first plane, beginning at the twelve o'clock position and moving in either the clockwise or anticlockwise direction. From the first plane move to the next position and continue numbering in the same direction (clockwise or anticlockwise), always returning to the twelve o'clock position or the position nearest to it in the forward direction before assigning locants in that plane. Continue numbering in this manner until all positions are numbered.

A full discussion of numbering deltahedra may be found elsewhere.⁸ The complete descriptor for the CSU should appear just before the central atom list. Where structurally significant, metal-metal bonds may be indicated (see Section IR-9.2.5.3 and examples below).

The chain or ring structure numbering in a CSU must be consecutive and only thereafter obey rules (a)–(d) given in Section IR 9.2.5.6. In Example 3 below, the CSU numbering in fact coincides with the numbering that would be reached using those rules alone.

Examples:

1. $[\{\text{Co}(\text{CO})_3\}_3(\mu_3\text{-CBr})]$
 $(\mu_3\text{-bromomethanetriido})\text{nonacarbonyl-}triangulo\text{-tricobalt}(3\text{ Co—Co})$, or
 $(\mu_3\text{-bromomethanetriido})\text{-}triangulo\text{-tris(tricarbonylcobalt)}(3\text{ Co—Co})$
2. $[\text{Cu}_4(\mu_3\text{-I})_4(\text{PEt}_3)_4]$
 $tetra\text{-}\mu_3\text{-iodido-tetrakis(triethylphosphane)-}tetrahedro\text{-tetracopper}$, or
 $tetra\text{-}\mu_3\text{-iodido-tetrakis(triethylphosphane)-}[T_d-(13)-\Delta^4-closo]\text{-tetracopper}$

3. $[\text{Co}_4(\text{CO})_{12}]$ 

tri- μ -carbonyl-1:2 κ^2 C;1:3 κ^2 C;2:3 κ^2 C-nonacarbonyl-
1 κ^2 C,2 κ^2 C,3 κ^2 C,4 κ^3 C-[T_d -(13)- Δ^4 -*closo*]-tetracobalt(6 Co—Co)

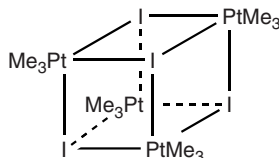
This compound has also been named, in Section II-5.3.3.3.6 of Ref. 7, using the chain and ring nomenclature (see Section IR-7.4). However, that name is based on a completely different numbering scheme.

Examples:

4. $[\text{Mo}_6\text{S}_8]^{2-}$

octa- μ_3 -sulfido-*octahedro*-hexamolybdate(2-), or
octa- μ_3 -sulfido-[O_h -(141)- Δ^8 -*closo*]-hexamolybdate(2-)

5.



tetra- μ_3 -iodido-1:2:3 κ^3 I;1:2:4 κ^3 I;1:3:4 κ^3 I;2:3:4 κ^3 I-dodecamethyl-
1 κ^3 C,2 κ^3 C,3 κ^3 C,4 κ^3 C-*tetrahedro*-tetraplatinum(IV), or
tetra- μ_3 -iodido-1:2:3 κ^3 I;1:2:4 κ^3 I;1:3:4 κ^3 I;2:3:4 κ^3 I-dodecamethyl-
1 κ^3 C,2 κ^3 C,3 κ^3 C,4 κ^3 C-[T_d -(13)- Δ^4 -*closo*]-tetraplatinum(IV)

6. $[(\text{HgMe})_4(\mu_4\text{-S})]^{2+}$

μ_4 -sulfido-tetrakis(methylmercury)(2+), or
tetramethyl-1 κ C,2 κ C,3 κ C,4 κ C- μ_4 -sulfido-*tetrahedro*-tetramercury(2+), or
tetramethyl-1 κ C,2 κ C,3 κ C,4 κ C- μ_4 -sulfido-[T_d -(13)- Δ^4 -*closo*]-tetramercury(2+)

IR-9.3 DESCRIBING THE CONFIGURATION OF COORDINATION ENTITIES

IR-9.3.1 Introduction

Once the constitution of a coordination entity has been defined, it remains to describe the spatial relationships between the structural components of the molecule or ion. Molecules

that differ only in the spatial distribution of the components are known as stereoisomers. Stereoisomers that are mirror images of one another are called enantiomers (sometimes these have been called optical isomers), while those that are not are called diastereoisomers (or geometrical isomers). This is an important distinction in chemistry as, in general, diastereoisomers exhibit different physical, chemical and spectroscopic properties from one another, while enantiomers exhibit identical properties (except in the presence of other chiral entities). It is instructive to consider an everyday analogy in order to establish how the configuration of a molecule (and the embedded spatial relationships) can be described.

Using the terminology introduced above, left and right hands may be regarded as enantiomers of one another, since they are different (non-superimposable), but they are mirror images of each other. In both cases the thumbs are adjacent to the index finger, and the components of each hand are similarly disposed relative to all the other parts of that hand. If the thumb and index finger of a right hand were to be exchanged, the resulting hand could be considered to be a diastereoisomer of the normal right hand (and it too would have an enantiomer, resulting from a similar exchange on a left hand). The key point is that the relative positions of the components of diastereoisomers (the normal right hand and the modified one) are different.

In order to describe the hand fully the components (four fingers, one thumb and the central part of the hand) must be identified, the points of attachment available on the hand, and the relative positions of the fingers and thumb around the hand, must be described and whether the hand is 'left' or 'right' must be specified. The last three steps deal with the configuration of the hand.

In the case of a coordination compound, the name and formula describe the ligands and central atom(s). Describing the configuration of such a coordination compound requires consideration of three factors:

- (i) coordination geometry – identification of the overall shape of the molecule;
- (ii) relative configuration – description of the relative positions of the components of the molecule, *i.e.* where the ligands are placed around the central atom(s) in the identified geometry;
- (iii) absolute configuration – identification of which enantiomer is being specified (if the mirror images are non-superimposable).

The next three sections deal with these steps in turn. A more detailed discussion of the configuration of coordination compounds can be found elsewhere.⁹

IR-9.3.2 Describing the coordination geometry

IR-9.3.2.1 *Polyhedral symbol*

Different geometrical arrangements of the atoms attached to the central atom are possible for all coordination numbers greater than one. Thus, two-coordinate species may involve a linear or a bent disposition of the ligands and central atom. Similarly, three-coordinate species may be trigonal planar or trigonal pyramidal, and four-coordinate species may be

square planar, square pyramidal or tetrahedral. The coordination polyhedron (or polygon in planar molecules) may be denoted in the name by an affix called the *polyhedral symbol*. This descriptor distinguishes isomers differing in the geometries of their coordination polyhedra.

The polyhedral symbol must be assigned before any other spatial features can be considered. It consists of one or more capital italic letters derived from common geometric terms which denote the idealized geometry of the ligands around the coordination centre, and an arabic numeral that is the coordination number of the central atom.

Distortions from idealized geometries commonly occur. However, it is normal practice to relate molecular structures to idealized models. The polyhedral symbol is used as an affix, enclosed in parentheses and separated from the name by a hyphen. The polyhedral symbols for the most common geometries for coordination numbers 2 to 9 are given in Table IR-9.2 and the corresponding structures and/or polyhedra are shown in Table IR-9.3.

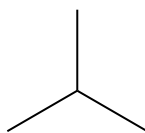
Table IR-9.2 *Polyhedral symbols*^a

<i>Coordination polyhedron</i>	<i>Coordination number</i>	<i>Polyhedral symbol</i>
linear	2	<i>L-2</i>
angular	2	<i>A-2</i>
trigonal plane	3	<i>TP-3</i>
trigonal pyramid	3	<i>TPY-3</i>
T-shape	3	<i>TS-3</i>
tetrahedron	4	<i>T-4</i>
square plane	4	<i>SP-4</i>
square pyramid	4	<i>SPY-4</i>
see-saw	4	<i>SS-4</i>
trigonal bipyramid	5	<i>TBPY-5</i>
square pyramid	5	<i>SPY-5</i>
octahedron	6	<i>OC-6</i>
trigonal prism	6	<i>TPR-6</i>
pentagonal bipyramid	7	<i>PBPY-7</i>
octahedron, face monocapped	7	<i>OCF-7</i>
trigonal prism, square-face monocapped	7	<i>TPRS-7</i>
cube	8	<i>CU-8</i>
square antiprism	8	<i>SAPR-8</i>
dodecahedron	8	<i>DD-8</i>
hexagonal bipyramid	8	<i>HBPY-8</i>
octahedron, <i>trans</i> -bicapped	8	<i>OCT-8</i>
trigonal prism, triangular-face bicapped	8	<i>TPRT-8</i>
trigonal prism, square-face bicapped	8	<i>TPRS-8</i>
trigonal prism, square-face tricapped	9	<i>TPRS-9</i>
heptagonal bipyramid	9	<i>HBPY-9</i>

^a Strictly, not all geometries can be represented by polyhedra.

Table IR-9.3 Polyhedral symbols, geometrical structures and/or polyhedra*Three-coordination*

trigonal plane



TP-3

trigonal pyramid



TPY-3

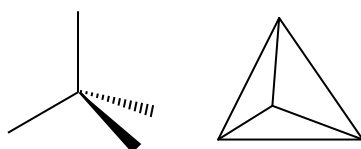
T-shape



TS-3

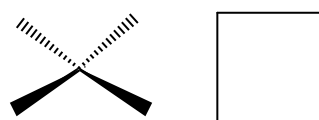
Four-coordination

tetrahedron



T-4

square plane



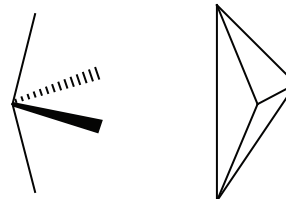
SP-4

square pyramid



SPY-4

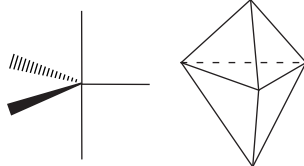
see-saw



SS-4

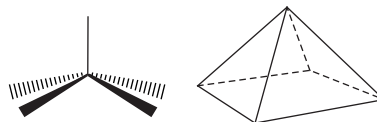
Five-coordination

trigonal bipyramid



TBPY-5

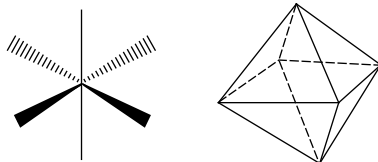
square pyramid



SPY-5

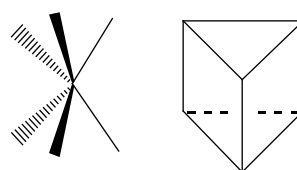
Six-coordination

octahedron



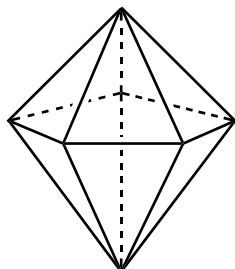
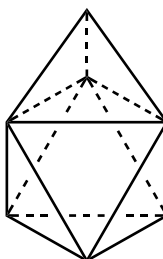
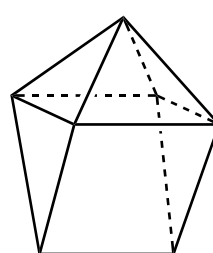
OC-6

trigonal prism

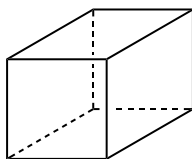
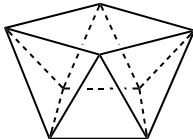


TPR-6

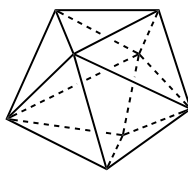
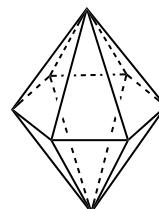
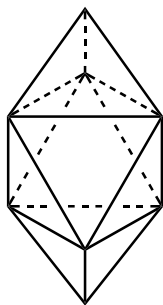
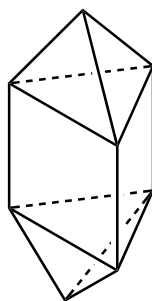
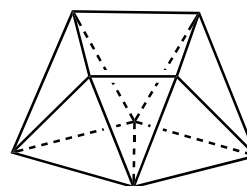
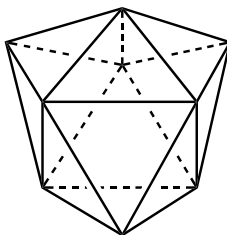
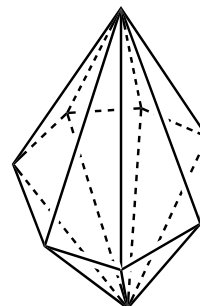
Table IR-9.3 Continued

*Seven-coordination*pentagonal
bipyramid*PBPY-7*octahedron, face
monocapped*OCF-7*trigonal prism,
square-face monocapped*TPRS-7**Eight-coordination*

cube

*CU-8*square
antiprism*SAPR-8*

dodecahedron

*DD-8*hexagonal
bipyramid*HBPY-8*octahedron,
trans-bicapped*OCT-8*trigonal prism,
triangular-face bicapped*TPRT-8*trigonal prism,
square-face bicapped*TPRS-8**Nine-coordination*trigonal prism,
square-face tricapped*TPRS-9*heptagonal
bipyramid*HBPY-9*

IR-9.3.2.2 *Choosing between closely related geometries*

For real molecules or ions, the stereochemical descriptor should be based on the nearest idealized geometry. However, some idealized geometries are closely related [*e.g.* square planar (*SP*-4), four-coordinate square pyramidal (*SPY*-4), see-saw (*SS*-4), and tetrahedral (*T*-4); T-shaped (*TS*-3), trigonal planar (*TP*-3), and trigonal pyramidal (*TPY*-3)] and care may therefore be required in making the choice.

The following approach is useful in determining the polyhedral symbol for four-coordinate structures. The key is to consider the locations of the central atom and the coordinating atoms in relation to each other. If all five atoms are in (or are close to being in) the same plane, then the molecule should be treated as square planar. If the four coordinating atoms are in a plane, but the central atom is significantly displaced from the plane, then the square pyramidal geometry is appropriate. If the four coordinating atoms do not lie in (or close to) a plane, then a polyhedron can be defined by joining all four coordinating atoms together with lines. If the central atom lies inside this polyhedron the molecule should be regarded as tetrahedral, otherwise, it should be regarded as having a see-saw structure.

T-shaped and trigonal planar molecules both have a central atom that lies in (or close to) the plane defined by the coordinating atoms. They differ in that the angles between the three coordinating atoms are approximately the same in the trigonal planar structure, while one angle is much larger than the other two in a T-shaped molecule. The central atom lies significantly out of the plane in a trigonal pyramidal structure.

IR-9.3.3 **Describing configuration – distinguishing between diastereoisomers**

IR-9.3.3.1 *General*

The placement of ligands around the central atom must be described in order to identify a particular diastereoisomer. There are a number of common terms (*e.g.* *cis*, *trans*, *mer* and *fac*) used to describe the relative locations of ligands in simple systems. However, they can be used only when a particular geometry is present (*e.g.* octahedral or square planar), and when there are only two kinds of donor atom present (*e.g.* Ma_2b_2 in a square planar complex, where M is a central atom and 'a' and 'b' are types of donor atom).

Several methods have been used to distinguish between diastereoisomers in more complex systems. Thus, stereoisomers resulting from the coordination of linear tetradentate ligands have often been identified as *trans*, *cis*- α , or *cis*- β ,¹⁰ and those resulting from coordination of macrocyclic tetradentate ligands have their own system.¹¹ The scope of most of these nomenclatures is generally quite limited, but a proposal with wider application in the description of complexes of polydentate ligands has been made more recently.¹²

Clearly a general method is required in order to distinguish between diastereoisomers of compounds in which either other geometries or more than two kinds of donor atoms are present. The *configuration index* has been developed for this purpose. The next section outlines the method by which a configuration index is obtained for a compound, and the

following sections give details for particular geometries. Commonly used terms are included for each geometry discussed.

IR-9.3.3.2 *Configuration index*

Once the coordination geometry has been specified by the polyhedral symbol, it becomes necessary to identify which ligands (or donor atoms) occupy particular coordination positions. This is achieved through the use of the configuration index which is a series of digits identifying the positions of the ligating atoms on the vertices of the coordination polyhedron. The configuration index has the property that it distinguishes between diastereoisomers. It appears within the parentheses enclosing the polyhedral symbol (see Section IR-9.3.2.1), following that symbol and separated from it by a hyphen.

Each donor atom must be assigned a priority number based on the rules developed by Cahn, Ingold and Prelog (the CIP rules).¹³ These priority numbers are then used to form the configuration index for the compound. The application of the CIP rules to coordination compounds is discussed in detail in Section IR-9.3.5 but, in general, donor atoms that have a higher atomic number have higher priority than those that have a lower atomic number.

The presence of polydentate ligands may require the use of primes on some of the numbers in the configuration index. The primes are used to indicate either that donor atoms are not part of the same polydentate ligand as those that have unprimed priority numbers, or that the donor atoms belong to different parts of a polydentate ligand that are related by symmetry. A primed priority number means that that donor atom has lower priority than the same kind of donor atom without a prime on the priority number. More detail on the 'priming convention' can be found in Section IR-9.3.5.3.

IR-9.3.3.3 *Square planar coordination systems (SP-4)*

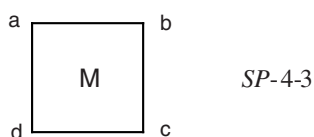
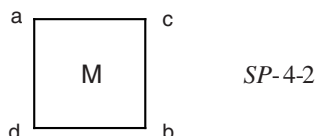
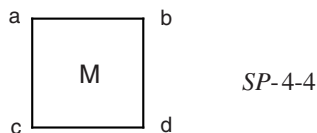
The terms *cis* and *trans* are used commonly as prefixes to distinguish between stereoisomers in square planar systems of the form $[\text{Ma}_2\text{b}_2]$, where M is the central atom, and 'a' and 'b' are different types of donor atom. Similar donor atoms occupy coordination sites adjacent to one another in the *cis* isomer, and opposite to one another in the *trans* isomer. The *cis-trans* terminology is not adequate to distinguish between the three isomers of a square planar coordination entity $[\text{Mabcd}]$, but could be used, in principle, for an $[\text{Ma}_2\text{bc}]$ system (where the terms *cis* and *trans* would refer to the relative locations of the similar donor atoms). This latter use is not recommended.

The configuration index for a square planar system is placed after the polyhedral symbol (SP-4). It is the single digit which is the priority number for the ligating atom *trans* to the ligating atom of priority number 1, *i.e.* the priority number of the ligating atom *trans* to the most preferred ligating atom.

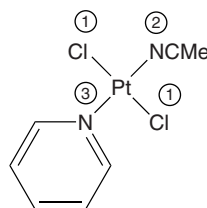
Examples:

1. Priority sequence: $a > b > c > d$

Priority number sequence: $1 < 2 < 3 < 4$



2.

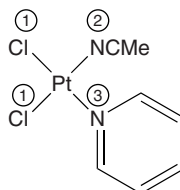


(SP-4-1)-(acetonitrile)dichlorido(pyridine)platinum(II)

If there are two possibilities, as in Example 3, the configuration index is the priority number with the higher numerical value. Both the priority 2 ligand (acetonitrile) and the priority 3 ligand (pyridine) are *trans* to a priority 1 ligand (chloride). The higher numerical value (3) is chosen for the configuration index. This choice is sometimes referred to as having been made according to the principle of *trans* maximum difference, *i.e.* that the difference between the numerical values of the priority numbers of the ligands should be as large as possible.

Example:

3.



(SP-4-3)-(acetonitrile)dichlorido(pyridine)platinum(II)

IR-9.3.3.4 Octahedral coordination systems (OC-6)

The terms *cis* and *trans* are used commonly as prefixes to distinguish between stereoisomers in octahedral systems of the form $[\text{Ma}_2\text{b}_4]$, where M is the central atom, and 'a' and 'b' are different types of donor atom, and in certain similar systems. The 'a' donors occupy adjacent coordination sites in the *cis* isomer, and opposite coordination sites in the *trans* isomer (Example 1).

The terms *mer* (meridional) and *fac* (facial) are used commonly to distinguish between stereoisomers of complexes of the form $[\text{Ma}_3\text{b}_3]$. In the *mer* isomer (Example 2) the two groups of three similar donors each lie on a meridian of the coordination octahedron, in planes that also contain the central atom. In the *fac* isomer (Example 3) the two groups of three similar donors each occupy coordination sites on the corners of a face of the coordination octahedron.

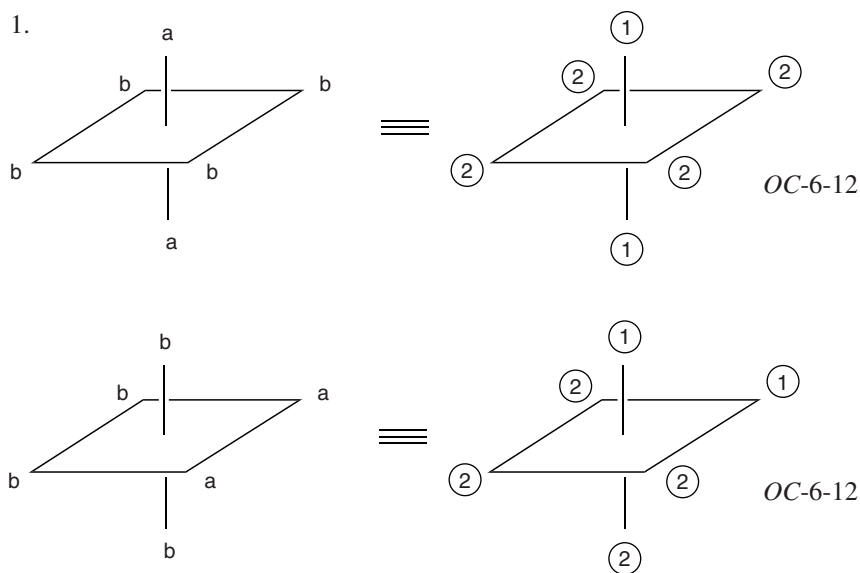
The configuration index of an octahedral system follows the polyhedral symbol (OC-6) and consists of two digits.

The first digit is the priority number of the ligating atom *trans* to the ligating atom of priority number 1, *i.e.* the priority number of the ligating atom *trans* to the most preferred ligating atom. If there is more than one ligating atom of priority 1, then the first digit is the priority number of the *trans* ligand with the highest numerical value (remembering that a primed number will be of higher numerical value than the corresponding unprimed number).

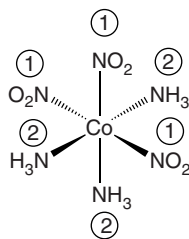
These two ligating atoms, the priority 1 atom and the (lowest priority) atom *trans* to it, define the *reference axis* of the octahedron.

The second digit of the configuration index is the priority number of the ligating atom *trans* to the most preferred ligating atom in the plane that is perpendicular to the reference axis. If there is more than one such ligating atom in that plane, the priority number of the *trans* atom having the largest numerical value is selected.

Examples:

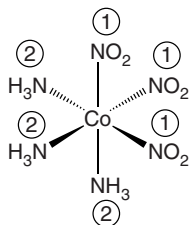


2.



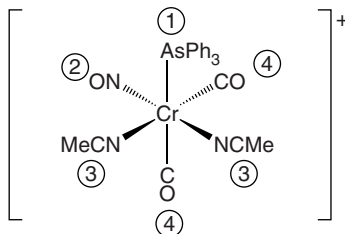
mer-[Co(NH₃)₃(NO₂)₃]
(OC-6-21)-triamminetrinitrito- κ^3N -cobalt(III)

3.



fac-[Co(NH₃)₃(NO₂)₃]
(OC-6-22)-triamminetrinitrito- κ^3N -cobalt(III)

4.



(OC-6-43)-bis(acetonitrile)dicarbonylnitrosyl(triphenylarsane)chromium(1+)

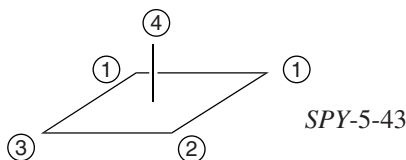
IR-9.3.3.5 Square pyramidal coordination systems (SPY-4, SPY-5)

The configuration index of an *SPY-5* system consists of two digits. The first digit is the priority number of the ligating atom on the C_4 symmetry axis (the reference axis) of the idealized pyramid. The second digit is the priority number of the ligating atom *trans* to the ligating atom with the lowest priority number in the plane perpendicular to the C_4 symmetry axis. If there is more than one such atom in the perpendicular plane, then the second digit is chosen to have the highest numerical value.

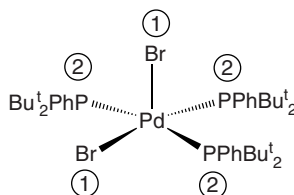
The configuration index of an *SPY-4* system is a single digit that is chosen in the same way as the second digit of *SPY-5* systems. The configuration index of a four-coordinate square pyramidal system will therefore be the same as that for the square planar structure that would result from the ligands and the central atom being coplanar. The difference between the structures is described by the polyhedral symbol rather than by the configuration index.

Examples:

1.



2.



(SPY-5-12)-dibromidotris[di-*tert*-butyl(phenyl)phosphane]palladium

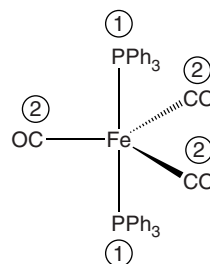
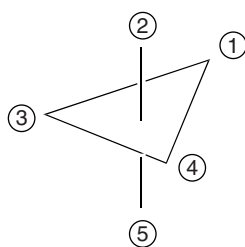
IR-9.3.3.6 Bipyramidal coordination systems (TBPY-5, PBPY-7, HBPY-8 and HBPY-9)

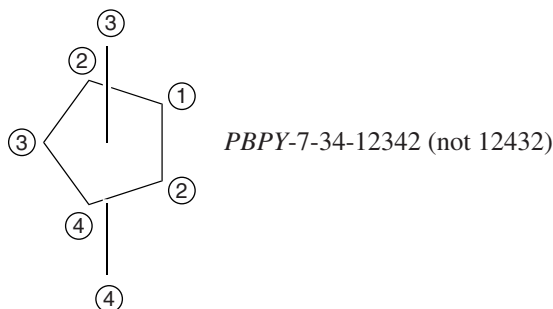
The configuration index for bipyramidal coordination systems follows the appropriate polyhedral symbol, and consists of two segments separated by a hyphen, except for the trigonal bipyramid where the second segment is not required and is therefore omitted. The first segment has two digits which are the priority numbers of the ligating atoms on the highest order rotational symmetry axis, the reference axis. The lower number is cited first.

The second segment consists of the priority numbers of the ligating atoms in the plane perpendicular to the reference axis. The first digit is the priority number for the preferred ligating atom, *i.e.* the lowest priority number in the plane. The remaining priority numbers are cited in sequential order proceeding around the projection of the structure either clockwise or anticlockwise, in whichever direction gives the lower numerical sequence. The lowest numerical sequence is that having the lower number at the first point of difference when the numbers are compared digit by digit from one end to the other.

Examples:

1. Trigonal bipyramid (TBPY-5)



2. Pentagonal bipyramid (*PBPY*-7)IR-9.3.3.7 *T-shaped systems (TS-3)*

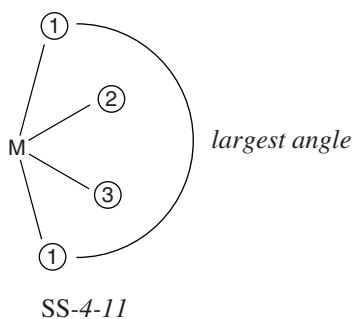
The configuration index for T-shaped systems follows the polyhedral symbol and consists of a single digit, the priority number of the ligating atom on the stem of the T (as opposed to the crosspiece of the T).

IR-9.3.3.8 *See-saw systems (SS-4)*

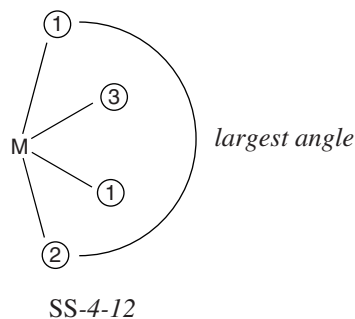
The configuration index for see-saw systems consists of two digits, the priority numbers of the two ligating atoms separated by the largest angle. The number of lower numerical value is cited first.

Examples:

1.



2.



IR-9.3.4

Describing absolute configuration – distinguishing between enantiomers

IR-9.3.4.1

General

There are two well-established, but fundamentally different, systems for distinguishing between two enantiomers (stereoisomers that are mirror images of one another). The first, based on the chemical constitution of the compound, involves the *R/S* convention used for describing tetrahedral centres and the closely related *C/A* convention used for other polyhedra. The *R/S* and *C/A* conventions use the priority sequence referred to in Section IR-9.3.3.2, and detailed in Section IR-9.3.5, where the ligating atoms are assigned a priority number based (usually) on their atomic number and their substituents.

The second is based on the geometry of the molecule and makes use of the skew-lines convention; it is usually applied only to octahedral complexes. The two enantiomers are identified by the symbols Δ and Λ in this system. The *C/A* nomenclature is not required for those chelate complexes where the skew-lines convention is completely unambiguous (see Sections IR-9.3.4.11 to 9.3.4.14).

IR-9.3.4.2 *The R/S convention for tetrahedral centres*

The convention used to describe the absolute configurations of tetrahedral centres was originally developed for carbon atom centres (see Ref. 13 and Section P-91 of Ref. 1) but can be used for any tetrahedral centre. There is no need to alter the rules in treating tetrahedral metal complexes.

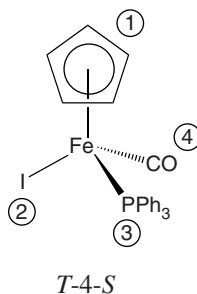
The symbol *R* is assigned if the cyclic sequence of priority numbers, proceeding from highest priority, is clockwise when the viewer is looking down the vector from the tetrahedral centre to the least preferred substituent (the substituent having the priority number with the highest numerical value, *i.e.* 4). An anticlockwise cyclic sequence is assigned the symbol *S*.



This system is most often used in conjunction with configuration internally in ligands but can be applied equally well to tetrahedral metal centres. It has also been useful for pseudotetrahedral organometallic complexes when, for example, cyclopentadienyl ligands are treated as if they were monodentate ligands of high priority.

Example:

1.

IR-9.3.4.3 *The R/S convention for trigonal pyramidal centres*

Molecules containing a trigonal pyramidal centre (TPY-3) may exist as a pair of stereoisomers. The configuration of this centre can be described in a similar way to that of a tetrahedral centre. This is achieved through notional placement of a 'phantom atom' of low priority in the coordination site that would create a tetrahedral centre from a trigonal pyramidal centre. The centre can then be identified as *R* or *S* by the methods described above.

The use of some bonding theories leads to the placement of a lone pair on a trigonal pyramidal centre. If this is done, the absolute configuration of the centre is also described by the *R/S* convention, in this case by placing the 'phantom atom' in the site that is occupied by the lone pair. Examples of this practice may be found in the description of absolute configurations for sulfoxides in which the alkyl substituents are different.

IR-9.3.4.4 *The C/A convention for other polyhedral centres*

The *R/S* convention makes use of priority numbers for the determination of chirality at tetrahedral centres, as detailed above. The same principles are readily extendable to geometries other than tetrahedral.¹⁴ However, in order to avoid confusion, and to emphasize the unique aspects of the priority sequence systems as applied to coordination polyhedra, the symbols *R* and *S* are replaced by the symbols *C* and *A* when applied to other polyhedra.

The procedure for arriving at ligating atom priorities is detailed in Section IR-9.3.5. Once these priorities have been assigned, the reference axis (and direction) appropriate to the geometry is identified. The priority numbers of the ligating atoms coordinated in the plane perpendicular to the reference axis are then considered, viewing from the axial ligating atom of higher priority.

Beginning with the highest priority atom in the plane perpendicular to the reference axis, the clockwise and anticlockwise sequences of priority numbers are compared, and that with the lower number at the first point of difference is chosen. If the chosen sequence results from a clockwise reading of the priority numbers, then the structure is given the chirality symbol *C*, otherwise it is given the symbol *A*.

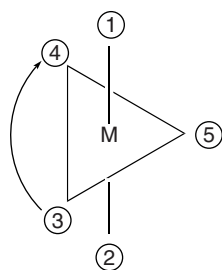
IR-9.3.4.5 *The C/A convention for trigonal bipyramidal centres*

The procedure is similar to that used for tetrahedral systems in the *R/S* convention, but it is modified because of the presence of a unique reference axis (running through the two axial donor atoms and the central atom).

The structure is oriented so that the viewer looks down the reference axis, with the more preferred donor atom (having a priority number with lower numerical value) closer to the viewer. Accordingly, the axial donor atom with the lower priority lies beyond the central atom. Using this orientation, the priority sequence of the three ligating atoms in the trigonal plane is examined. If the sequence proceeds from the highest priority to the lowest priority in a clockwise fashion, the chirality symbol *C* is assigned. Conversely, if the sequence from highest to lowest priority (from lowest numerical index to highest numerical index) is anticlockwise, the symbol *A* is assigned.

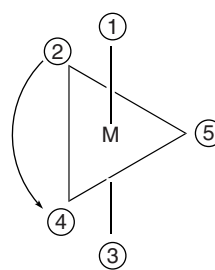
Examples:

1.



Chirality symbol = *C*

2.



Chirality symbol = *A*

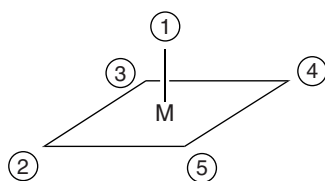
IR-9.3.4.6 *The C/A convention for square pyramidal centres*

A procedure similar to that described in Section IR-9.3.4.4 is used for square pyramidal structures. In the case of *SPY-5* systems, the polyhedron is oriented so that the viewer looks along the formal C_4 axis, from the axial ligand toward the central atom. The priority numbers of the ligating atoms in the perpendicular plane are then considered, beginning with the highest priority atom (the one having the priority number of lowest numerical value). The clockwise and anticlockwise sequences of priority numbers are compared, and the structure is assigned the symbol *C* or *A* according to whether the clockwise (*C*) or anticlockwise (*A*) sequence is lower at the first point of difference.

The chirality of an *SPY-4* system is defined in a similar way. In this case, the viewer looks along the formal C_4 axis in such a way that the ligands are further away than the central atom. The priority numbers are then used to assign the symbol *C* or *A*, as for the *SPY-5* system.

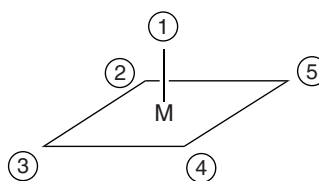
Examples:

1.



Chirality symbol = *C*

2.



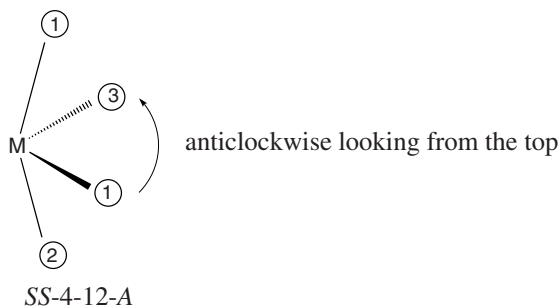
Chirality symbol = *A*

IR-9.3.4.7 *The C/A convention for see-saw centres*

The absolute configurations of see-saw complexes can be described using the *C/A* system. The configuration index for see-saw systems consists of two digits, the priority numbers of the two ligands separated by the largest angle. The higher priority ligand of these two is identified and used as a point from which to view the two ligands not involved in the configuration index. If moving from the higher priority ligand to the lower (through the smaller angle) entails making a clockwise motion, the absolute configuration is assigned *C*. An anticlockwise direction results in the absolute configuration *A*.

Example:

1.

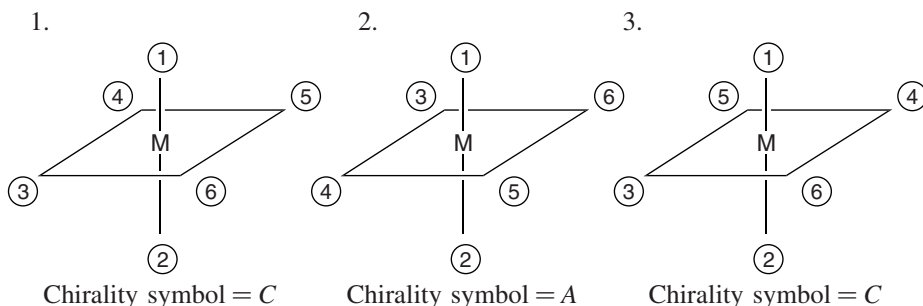


IR-9.3.4.8 *The C/A convention for octahedral centres*

The absolute configurations of some octahedral complexes can be described using either the skew-line reference system (Section IR-9.3.4.11) or the *C/A* system. The first is used more commonly, but the *C/A* system is more general and may be used for most complexes. The skew-line reference system is only applicable to tris(bidentate), bis(bidentate) and closely related systems.

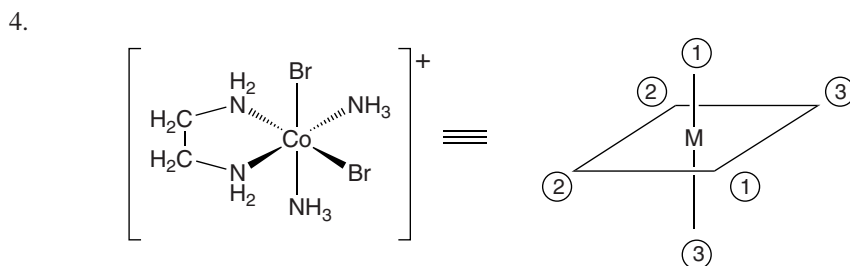
The reference axis for an octahedral centre is that axis containing the ligating atom of CIP priority 1 and the *trans* ligating atom of lowest possible priority (highest numerical value) (see Section IR-9.3.3.4). The atoms in the coordination plane perpendicular to the reference axis are viewed from the ligating atom having that highest priority (CIP priority 1) and the clockwise and anticlockwise sequences of priority numbers are compared. The structure is assigned the symbol *C* or *A*, according to whether the clockwise (*C*) or anticlockwise (*A*) sequence is lower at the first point of difference.

Examples:



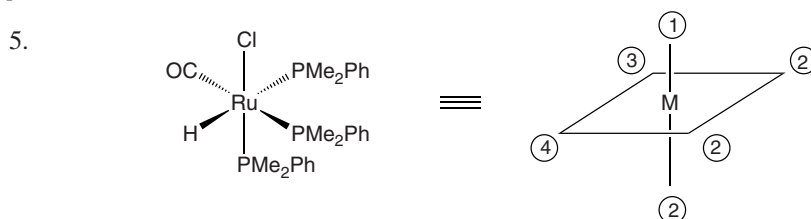
Example 4 shows the compound $[\text{CoBr}_2(\text{en})(\text{NH}_3)_2]^+$ which has the polyhedral symbol *OC*-6 and the configuration index 32. The chirality symbol is *C*.

Example:



Example 5 shows the complex $[\text{Ru}(\text{CO})\text{ClH}(\text{PMe}_2\text{Ph})_3]$ which has the descriptor *OC*-6-24-*A*. The chloride ligand has priority 1.

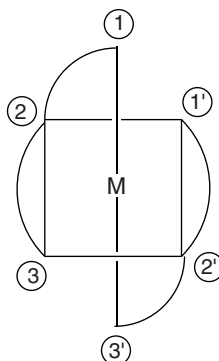
Example:



The *C/A* assignment for polydentate ligands is illustrated by Example 6 which uses the priming convention developed in Section IR-9.3.5. Note that priority number 2 has higher priority than 2'.

Example:

6.



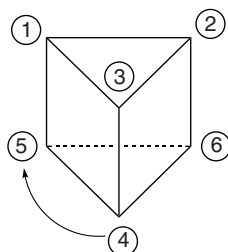
Chirality symbol = A

IR-9.3.4.9 *The C/A convention for trigonal prismatic centres*

For the trigonal prismatic system, the configuration index is derived from the CIP priority numbers of the ligating atoms opposite the triangular face containing the greater number of ligating atoms of highest CIP priority. The chirality symbol is assigned by viewing the trigonal prism from above the preferred triangular face and noting the direction of progression of the priority sequence for the less preferred triangular face.

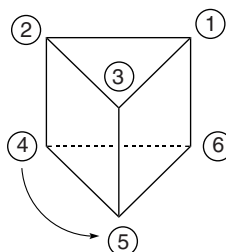
Examples:

1.



Chirality symbol = C

2.

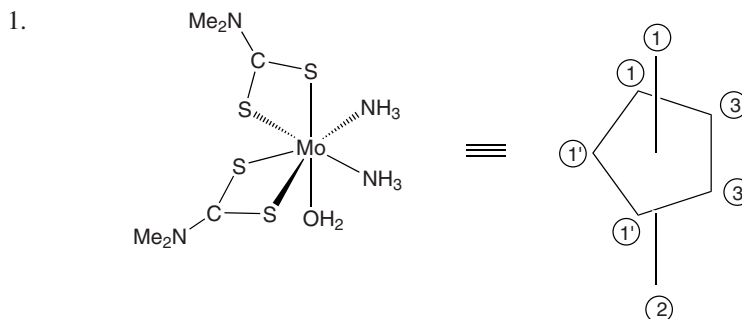


Chirality symbol = A

IR-9.3.4.10 *The C/A convention for other bipyramidal centres*

The procedure used for the trigonal bipyramid is appropriate for other bipyramidal structures. The structure is assigned the symbol *C* or *A*, according to whether the clockwise (*C*) or anticlockwise (*A*) sequence is lower at the first point of difference when the numbers are compared digit by digit from one end to the other (see Sections IR-9.3.4.5 and IR-9.3.4.6) and the molecule is viewed from the higher priority ligating atom on the reference axis.

Example:



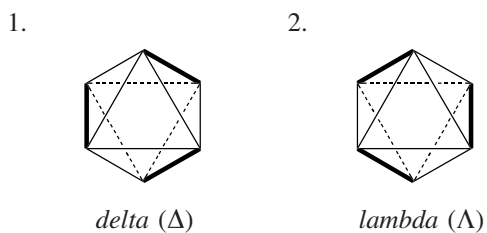
PBPY-7-12-11'1'33-A

IR-9.3.4.11 The skew-lines convention

Tris(bidentate) complexes constitute a general family for which a useful, unambiguous convention has been developed based on the orientation of skew lines which define a helix.

Examples 1 and 2 represent the *delta* (Δ) and *lambda* (Λ) forms of a complex such as $[\text{Co}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{3+}$. The rules define the chiralities of two additional families of structures. These are the *cis*-bis(bidentate) octahedral structures and the conformations of certain chelate rings. It is possible to use the system described below for complexes of higher polydentate ligands, but additional rules are required.¹⁵

Examples:



Two skew-lines which are not orthogonal possess the property of having one, and only one, normal in common. They define a helical system, as illustrated in Figures IR-9.1 and IR-9.2 (below). In Figure IR-9.1, one of the skew-lines, AA, determines the axis of a helix upon a cylinder whose radius is equal to the length of the common normal, NN, to the two skew-lines, AA and BB. The other of the skew-lines, BB, is a tangent to the helix at N and determines the pitch of the helix. In Figure IR-9.2, the two skew-lines AA and BB are seen in projection onto a plane orthogonal to their common normal.

Parts (a) of Figures IR-9.1 and IR-9.2 illustrate a right-handed helix to be associated with the Greek letter delta (Δ referring to configuration, δ to conformation). Parts (b) of Figures IR-9.1 and IR-9.2 illustrate a left-handed helix to be associated with the Greek letter lambda (Λ for configuration, λ for conformation). In view of the symmetry of the representation constituted by two skew-lines, the helix which the first line, say BB, determines around the second, AA, has the same chirality as that which AA determines around BB. As one of the lines is rotated about NN with respect to the other, inversion occurs when the lines are parallel or perpendicular (Figure IR-9.1).

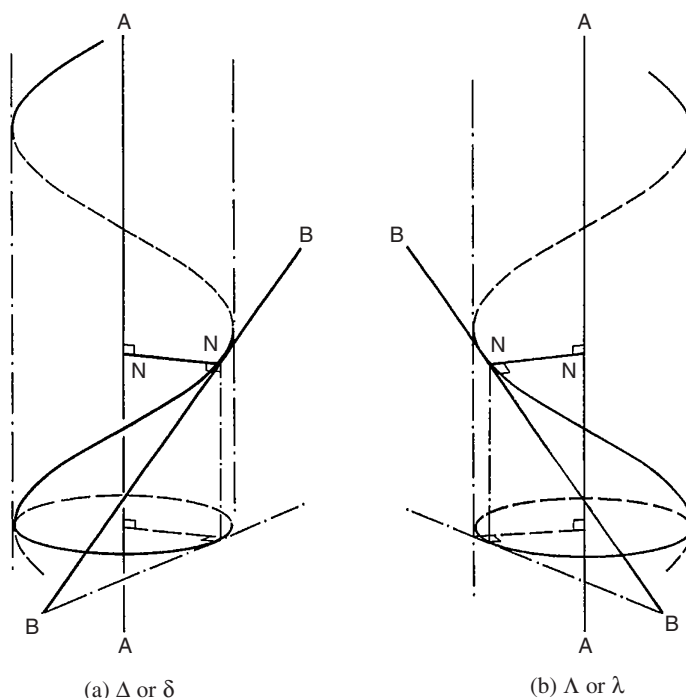


Figure IR-9.1. Two skew lines AA and BB which are not orthogonal define a helical system. In the Figure, AA is taken as the axis of a cylinder whose radius is determined by the common normal NN of the two skew-lines. The line BB is a tangent to the above cylinder at its crossing point with NN and defines a helix upon this cylinder. Cases (a) and (b) illustrate a right- and left-handed helix, respectively.

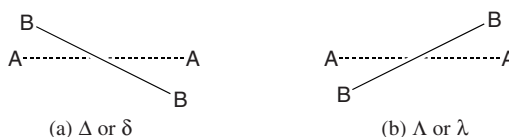


Figure IR-9.2. The figure shows pairs of non-orthogonal skew-lines in projection upon a plane parallel to both lines. The full line BB is above the plane of the paper, the dotted line AA is below this plane. Case (a) corresponds to (a) of Figure IR-9.1 and defines a right-handed helix. Case (b) corresponds to (b) of Figure IR-9.1 and defines a left-handed helix.

IR-9.3.4.12 Application of the skew-lines convention to tris(bidentate) octahedral complexes

Any two of the three chelate rings may be chosen to designate the configuration of tris(bidentate) coordination compounds. The donor atoms of each chelate ring define a line. Two such lines for a pair of chelate rings in the same complex define a helix, one line being the axis of the helix and the other a tangent to the helix at the normal common to the skew-lines. The tangent describes a right-handed (Δ) or a left-handed (Λ) helix with respect to the axis and thereby defines the chirality of that configuration.

IR-9.3.4.13 Application of the skew-lines convention to bis(bidentate) octahedral complexes

Figure IR-9.3(a) shows a common orientation of an octahedral tris(bidentate) structure projected onto a plane orthogonal to the three-fold axis of the structure. Figure IR-9.3(b)

shows the same structure oriented to emphasize the skew-line relationship between a pair of chelate rings that can be used to define chirality. Figure IR-9.3(c) shows that the same convention can be used for the *cis*-bis(bidentate) complex. The two chelate rings define the two skew-lines that, in turn, define the helix and the chirality of the substance. The procedure is precisely the same as that described for the tris(bidentate) case, but only a single pair of chelate rings is available.

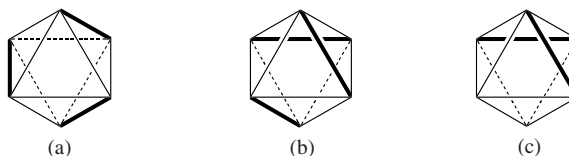


Figure IR-9.3. Two orientations of a tris(bidentate) structure, (a) and (b), to show the chiral relationship between these two species and the bis(bidentate) structure (c).

IR-9.3.4.14 Application of the skew-lines convention to conformations of chelate rings

In order to assign the chirality of a ring conformation, the line AA in Figure IR-9.2 is defined as that line joining the two ligating atoms of the chelate ring. The other line BB is that joining the two ring atoms which are neighbours to each of the ligating atoms. These two skew-lines define a helix in the usual way. The tangent describes a right-handed (δ) or a left-handed (λ) helix with respect to the axis and thereby defines the conformation in terms of the convention given in Figure IR-9.1. The relationship between the convention of Figure IR-9.2 and the usual representation of chelate ring conformation may be seen by comparing Figures IR-9.2 and IR-9.4.

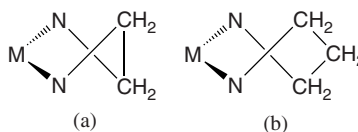


Figure IR-9.4. δ -Conformation of chelate rings: (a) five-membered; (b) six-membered.

IR-9.3.5 Determining ligand priority

IR-9.3.5.1 General

The methods for differentiating between stereoisomers outlined earlier in this Chapter require the assignment of priorities for the ligand atoms attached to the central atom (*i.e.* the donor atoms). These priority numbers are then used in the configuration index, which describes the relative positions of the ligands, and in the assignment of the absolute configuration of the compound.

The following sections outline the method used to arrive at the priority numbers for a given set of donor atoms, and the ways that the basic rules have to be modified in order to describe adequately systems that include polydentate ligands. These modifications, which are collectively referred to as the priming convention, make use of primes on the priority numbers to indicate which donor atoms are grouped together within a particular polydentate ligand.

IR-9.3.5.2 *Priority numbers*

The procedure for assigning priorities in mononuclear coordination systems is based on the standard sequence rules developed for chiral carbon compounds by Cahn, Ingold and Prelog.¹³ (See also Section P-91 of Ref. 1.) These CIP rules can be used quite generally for assigning priorities to groups attached to a central atom.

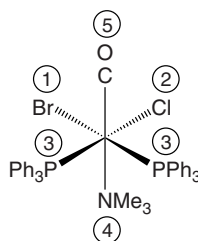
The essence of these rules, when applied to coordination compounds, is that the ligands attached to the central atom are compared to one another, beginning with the donor atom and then moving outwards in the structure. The comparison is made on the basis of atomic number and then, if required (*e.g.* when isotopes are being specified), atomic mass. Other properties may be used for subsequent comparisons, but the need for them is sufficiently rare that they need not be detailed here.

Once the ligands have been compared, the priority numbers are assigned as follows:

- (i) identical ligands are assigned the same priority,
- (ii) the ligand(s) with highest priority is (are) assigned the priority number 1; those with the next highest priority, 2; and so on.

Examples:

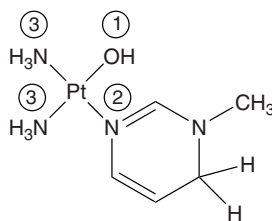
1.



Priority sequence: $\text{Br} > \text{Cl} > \text{PPh}_3, \text{PPh}_3 > \text{NMe}_3 > \text{CO}$

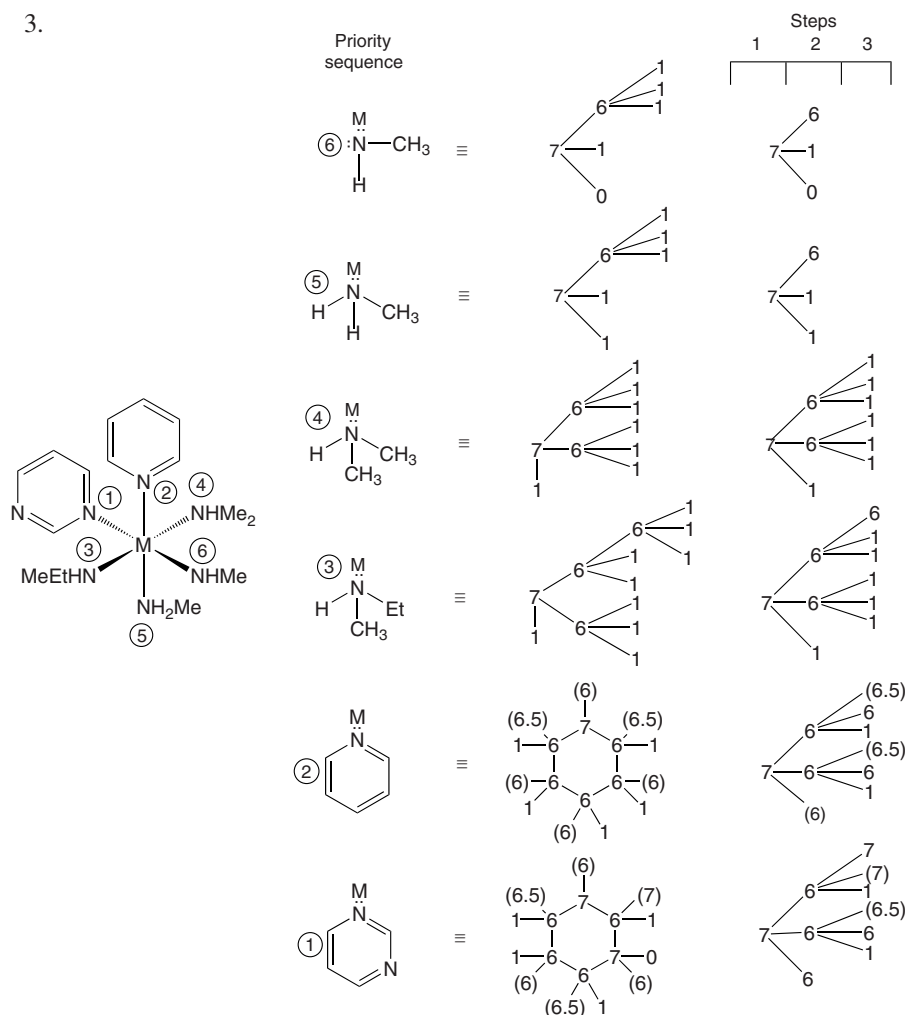
Priority numbers sequence: $1 > 2 > 3, 3 > 4 > 5$

2.



In Example 2, the heterocyclic ligand is given priority 2 since it has a lower atomic number donor atom than OH, and the substitution of the nitrogen donor ranks it above the ammine ligands.

3.



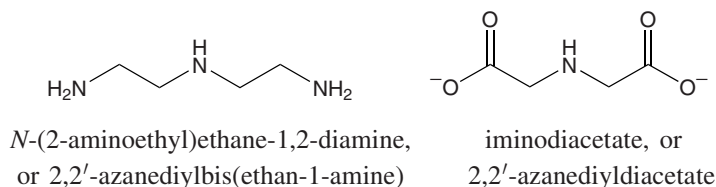
In Example 3, all the ligating atoms are nitrogen atoms. The key illustrates how proceeding along the branches of the ligand constituents allows priorities to be assigned. The numbers in columns 1, 2 and 3 on the right are the atomic numbers of the atoms in the structures, with those in brackets being used to take account of the presence of multiple bonds. The averaging techniques used in the case of resonance structures (last two ligands in the list) are given in the original paper.¹³

IR-9.3.5.3

Priming convention

The priming convention is required in order to avoid ambiguity when using the configuration index to describe the stereochemistry of systems that contain either more than one polydentate ligand of a particular kind, or a polydentate ligand that contains more than one coordinating fragment of a particular kind. This situation is found commonly with bis(tridentate) complexes, but also arises in more complicated cases. The need for this convention is best illustrated by example.

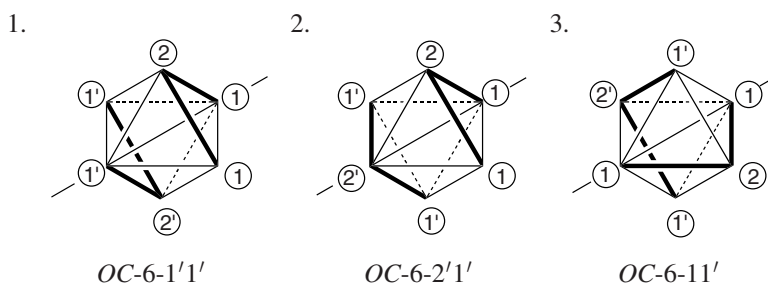
Bis(tridentate) complexes (*i.e.* octahedral complexes containing two identical linear tridentate ligands) may exist in three stereoisomeric forms, and there will be more if the tridentate ligands do not themselves contain some symmetry elements. The three isomers of the simplest case are represented below (Examples 1, 2 and 3), along with their polyhedral symbols (Section IR-9.3.2.1) and configuration indexes (Section IR-9.3.3.4). Complexes of *N*-(2-aminoethyl)ethane-1,2-diamine and iminodiacetate can be described by these diagrams.



The need for the priming convention can be seen by considering what the configuration indexes of Examples 1 and 3 would be in the absence of the priming convention. The two ligands are identical and consist of two similar fragments fused together. If the primes are ignored, the two complexes have the same distributions of ligating atoms (four donors of priority 1 in a square plane, and two of priority 2 *trans* to one another). They would therefore have the same configuration index, even though they are clearly different complexes.

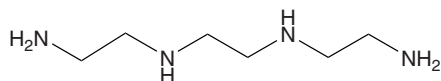
One way to highlight the difference between these two examples is to note that, in Example 1, all the donor atoms are *trans* to donors that are part of the other ligand. This is not true in Example 3. Using primes to indicate the groupings of donor atoms in particular ligands allows these two stereoisomers to be distinguished from one another by their configuration indexes.

Examples:

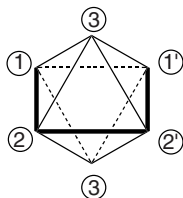


The priority numbers on one of the ligands are arbitrarily primed. The primed number is assigned lower priority than the corresponding unprimed number, but a higher priority than the next higher unprimed number. Thus 1' has lower priority than 1, but higher than 2.

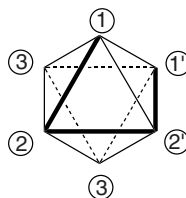
The technique also distinguishes between stereoisomers for complexes of higher polydentate ligands as indicated in Examples 4, 5 and 6 for linear tetradentate ligands such as *N,N'*-bis(2-aminoethyl)ethane-1,2-diamine. In this case, the donor atom priority numbers in half of the tetradentate ligand have been primed.

*N,N'*-bis(2-aminoethyl)ethane-1,2-diamine*Examples:*

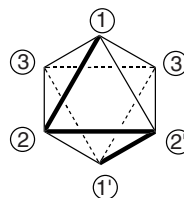
4.

*OC-6-2'2*

5.

*OC-6-32*

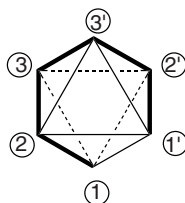
6.

*OC-6-1'3*

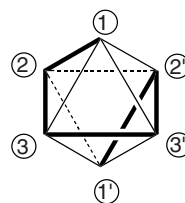
Pentadentate and hexadentate ligands can be treated similarly. Examples 7 and 8 apply to stereoisomers of classic linear hexadentate ligands, whereas Examples 9 and 10 apply to ligands containing a branched structure.

Examples:

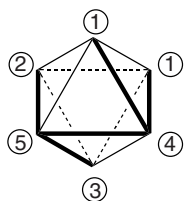
7.

*OC-6-3'3*

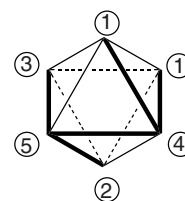
8.

*OC-6-1'3'*

9.

*OC-6-53*

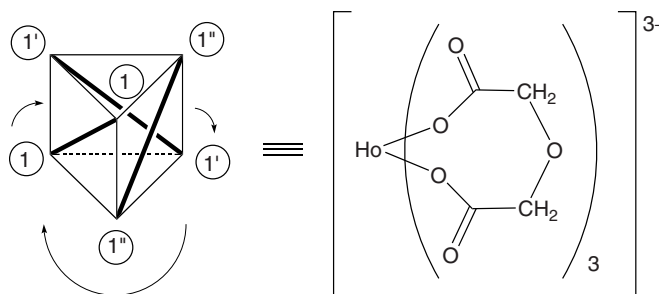
10.

*OC-6-52*

Example 11 illustrates the use of priming for assigning absolute configuration in a non-octahedral structure. The chirality designation is determined by the system of assigning primes to ligands with the extra requirement that the symbol 1 on the top face is placed above the symbol 1'' on the bottom face. This produces the sequence shown and the chirality symbol *C* when viewing the prism from above the top face. The stereochemical descriptor is *TPR-6-1''11'-C*. The skew-lines method (Section I-9.3.4.11) could also be applied and would give in this case the descriptor Δ .

Example:

11.



IR-9.4

FINAL REMARKS

This chapter has described the means by which coordination compounds can be named and formulated. These processes involve firstly identifying the central atom(s) and ligands (by name, formula or abbreviation, depending on the context), and secondly defining the nature of the attachment between the central atom(s) and the ligands. This latter step requires that the coordinating atoms in the ligand be identified (if there is any ambiguity), and that the spatial relationships between the ligands be described. The spatial relationships between the ligands are defined in terms of the coordination polyhedron (giving the polyhedral symbol) and the CIP priorities of the donor atoms (giving the configuration index and the absolute configuration).

IR-9.5

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2. In *Nomenclature of Inorganic Chemistry, IUPAC Recommendations 1990*, ed. G.J. Leigh, Blackwell Scientific Publications, Oxford, 1990, the term didentate was used rather than bidentate, for reasons of linguistic consistency. Reversion to the previously accepted term, bidentate, reflects common usage.
3. The names of the other hydrogen isotopes are discussed in Section IR-3.3.2.
4. The names of organic ligands should be assigned in accordance with IUPAC recommendations, see Ref. 1.
5. In order to simplify the rules and to resolve ambiguities that may arise when it is not clear whether a ligand is charged or not, the charge on a ligand is no longer considered in determining ligand order in the formulae of coordination compounds. (In *Nomenclature of Inorganic Chemistry, IUPAC Recommendations 1990*, ed. G.J. Leigh, Blackwell Scientific Publications, Oxford, 1990, anionic ligands were listed before neutral ligands.)
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